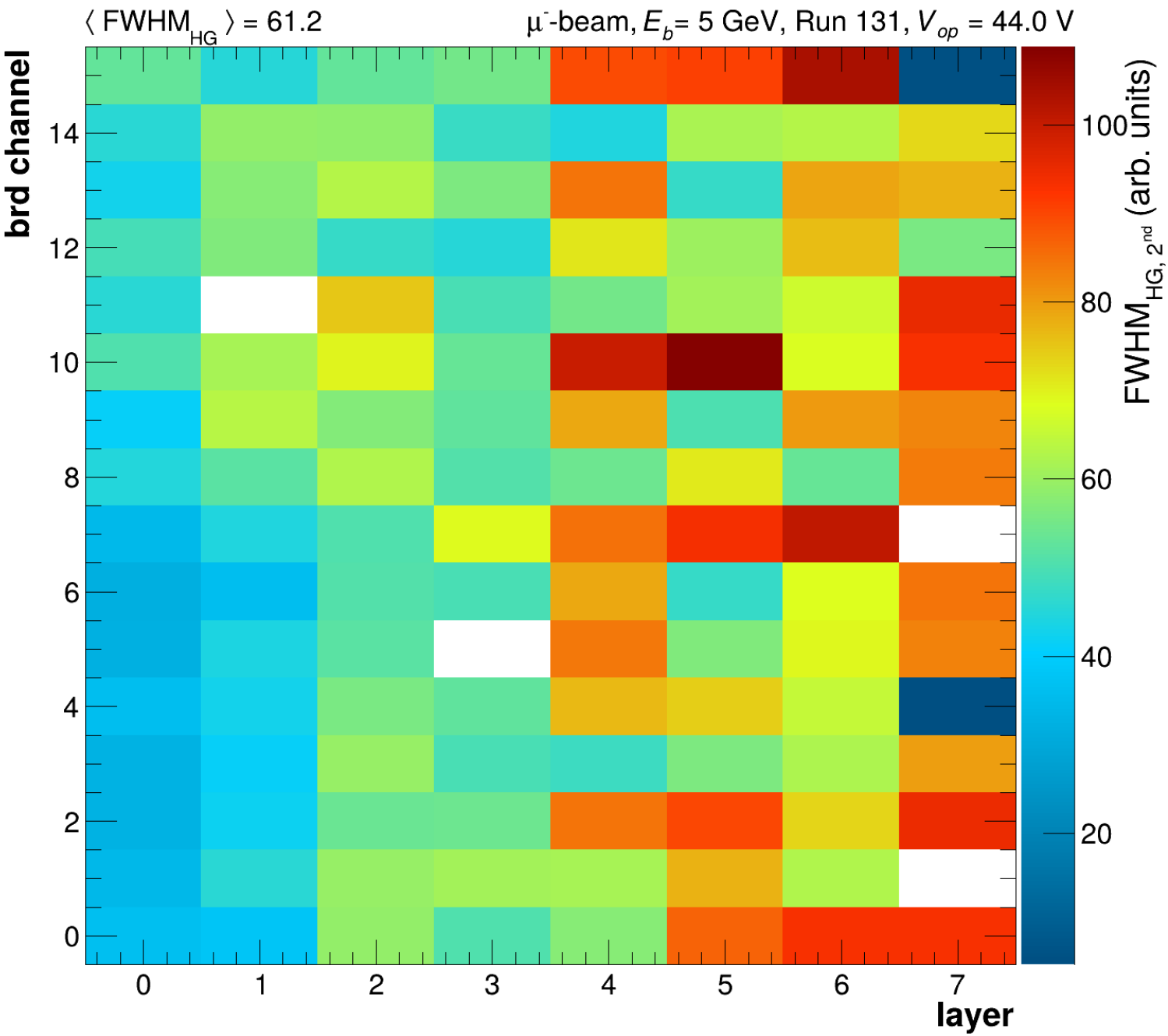
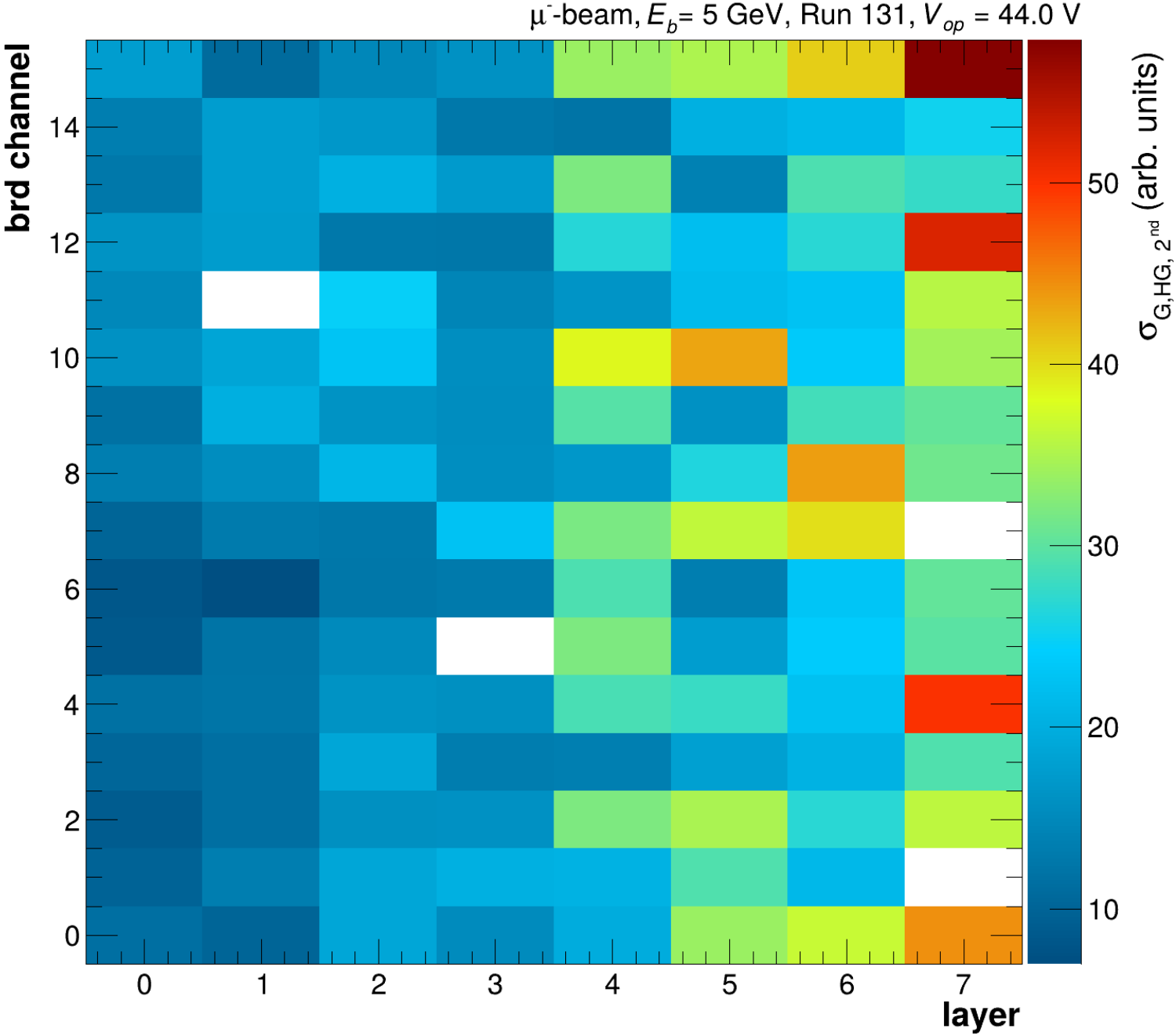
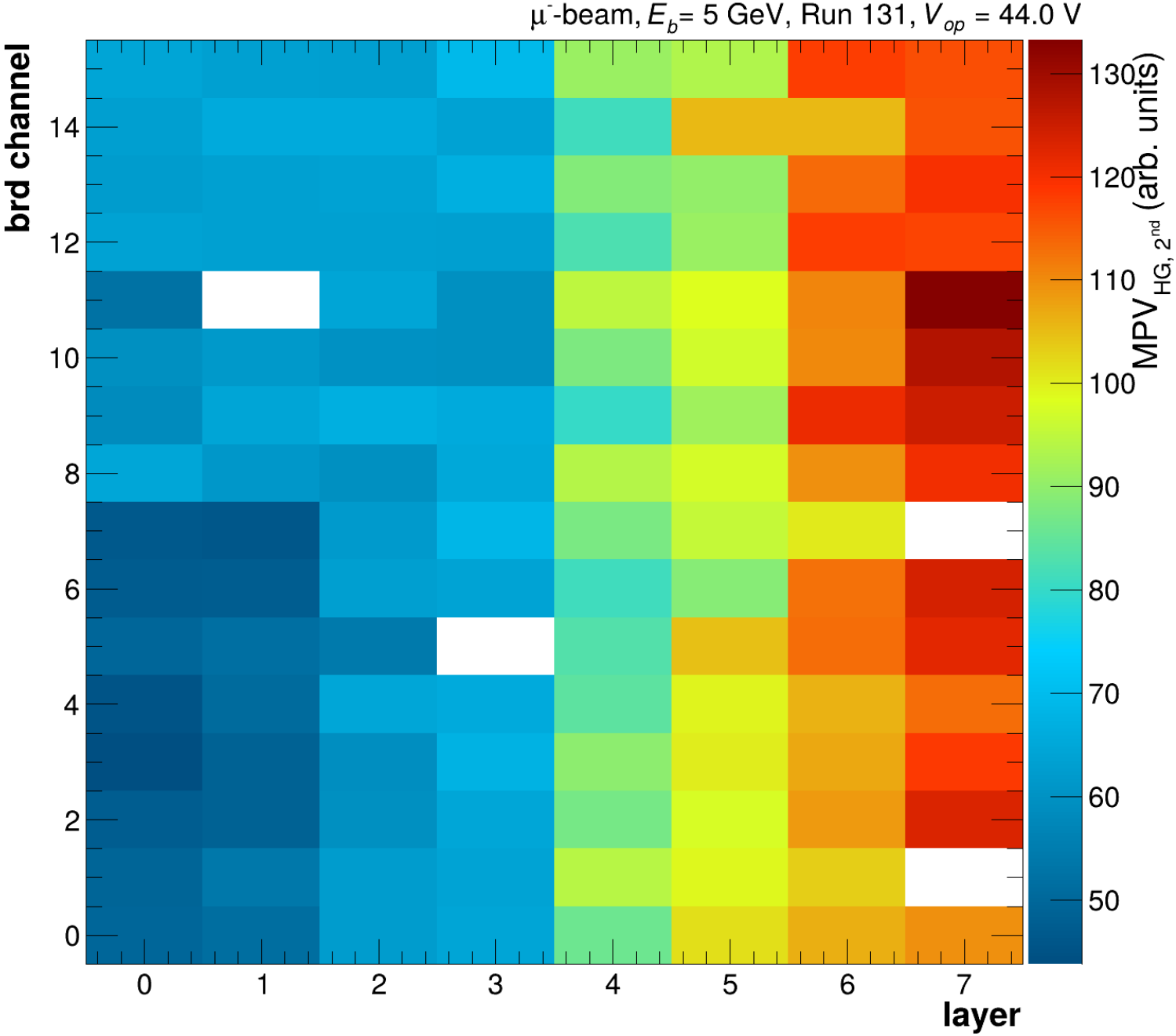
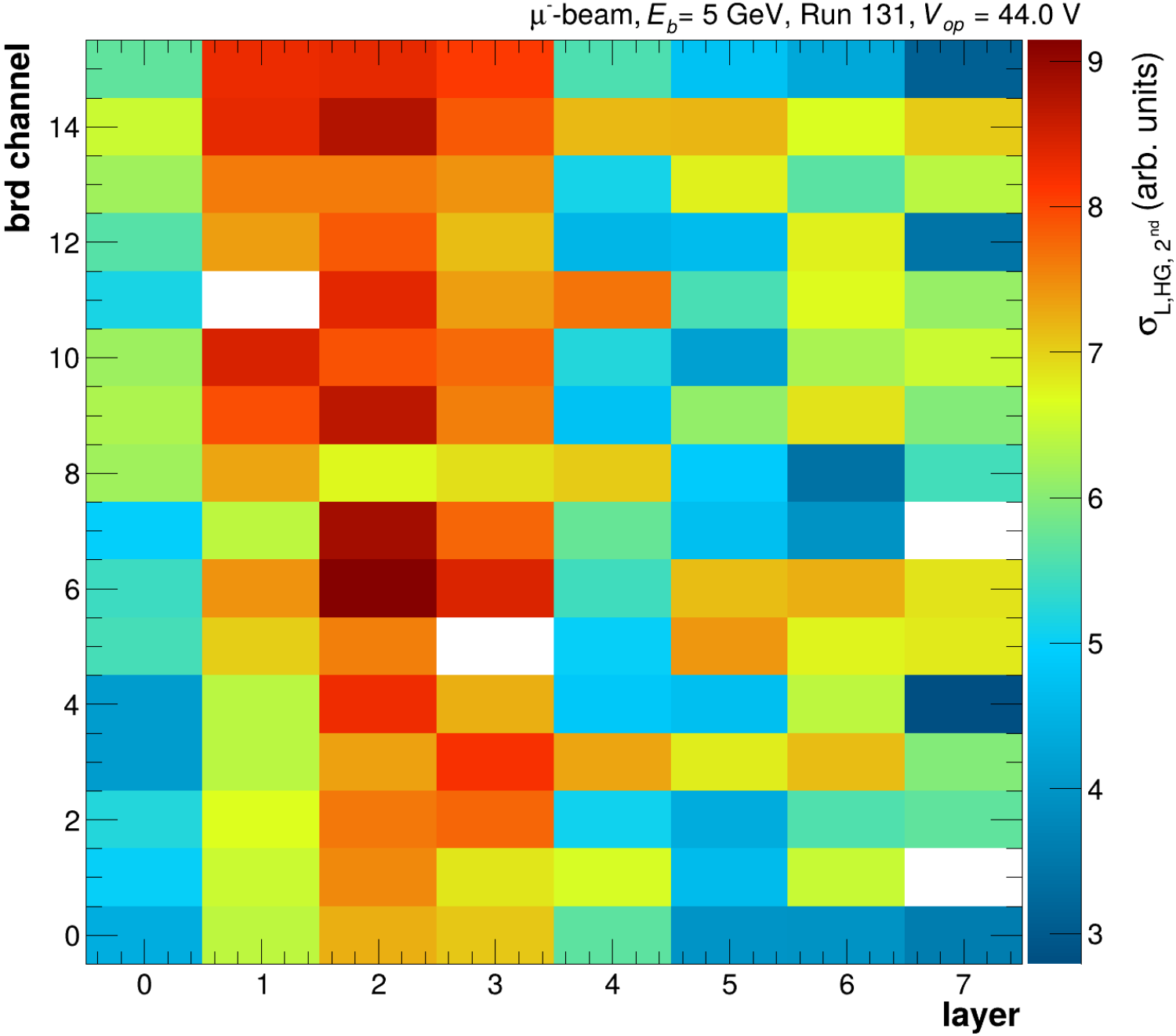
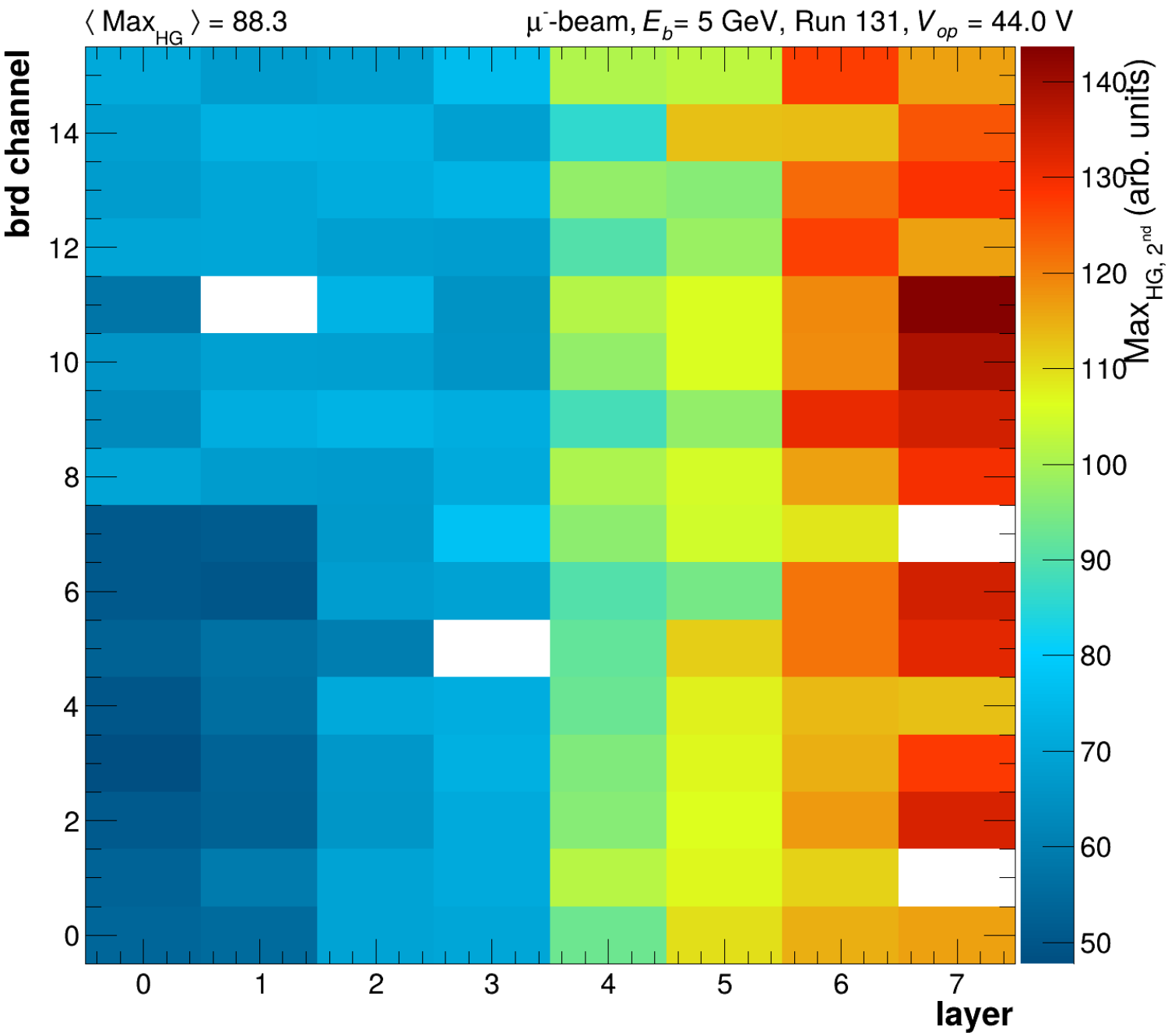
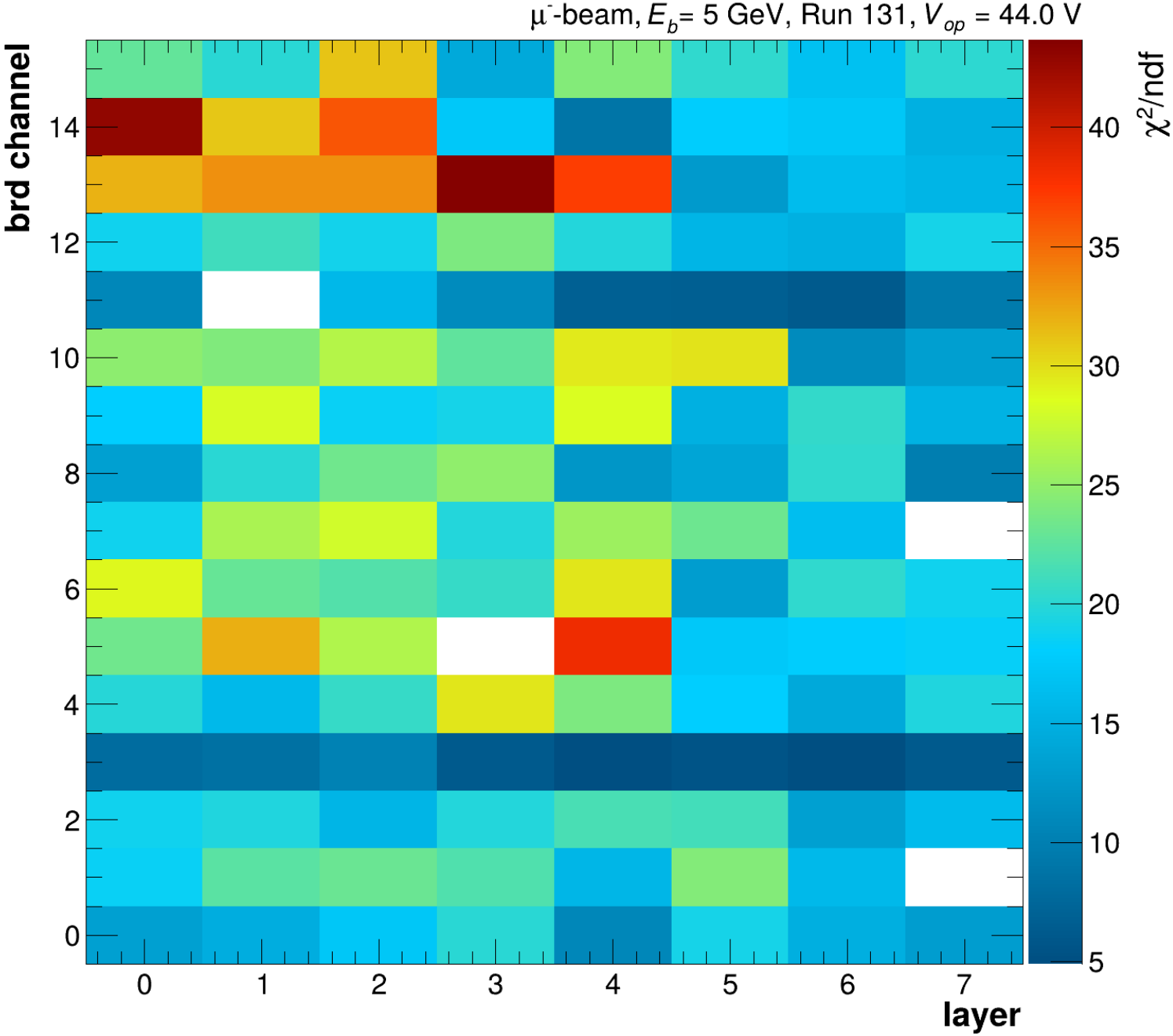

⟨ FWHM_HG ⟩ = 61.2
μ⁻-beam, E_b = 5 GeV, Run 131, V_op = 44.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
40
60
80
100

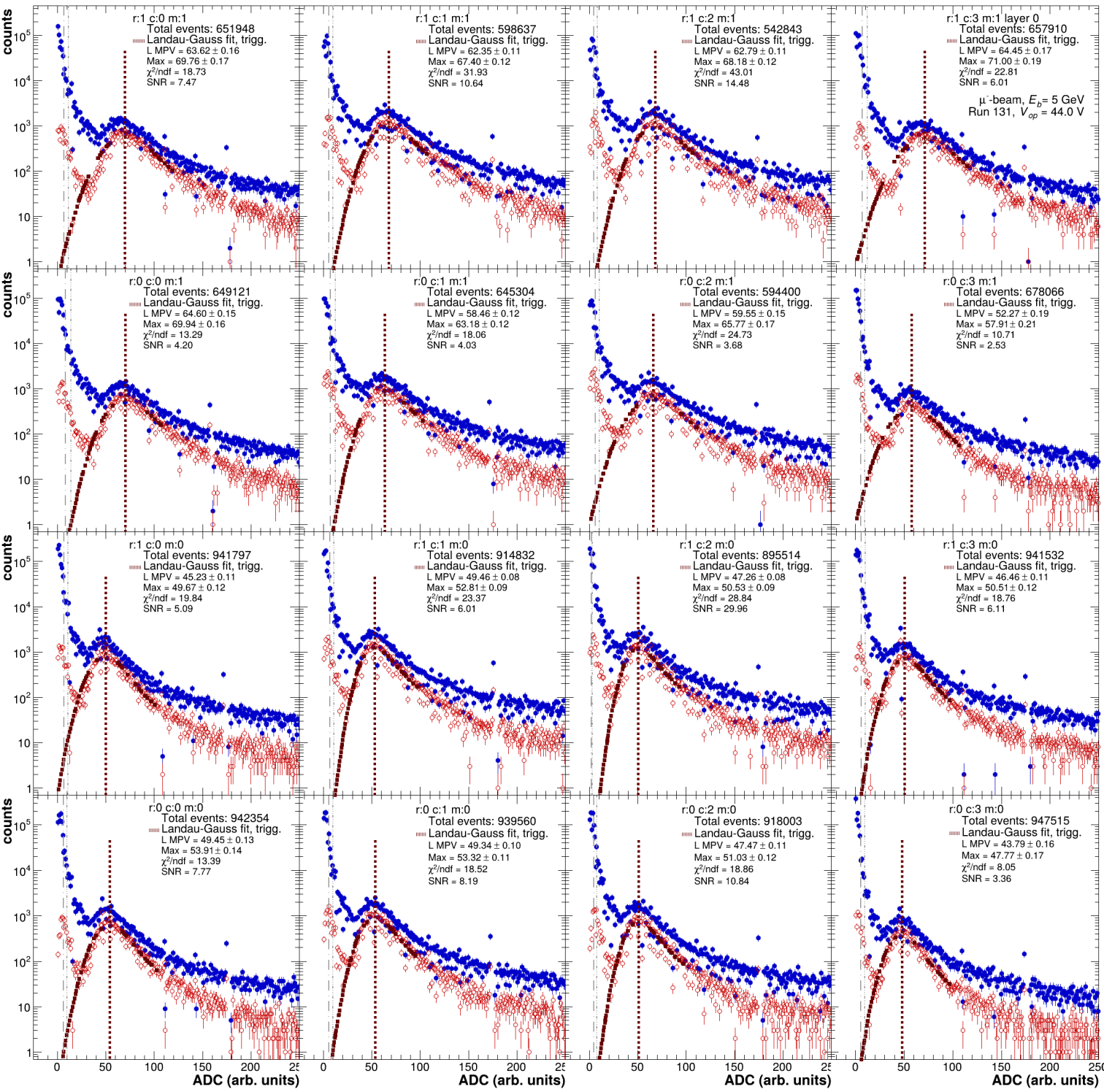

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50

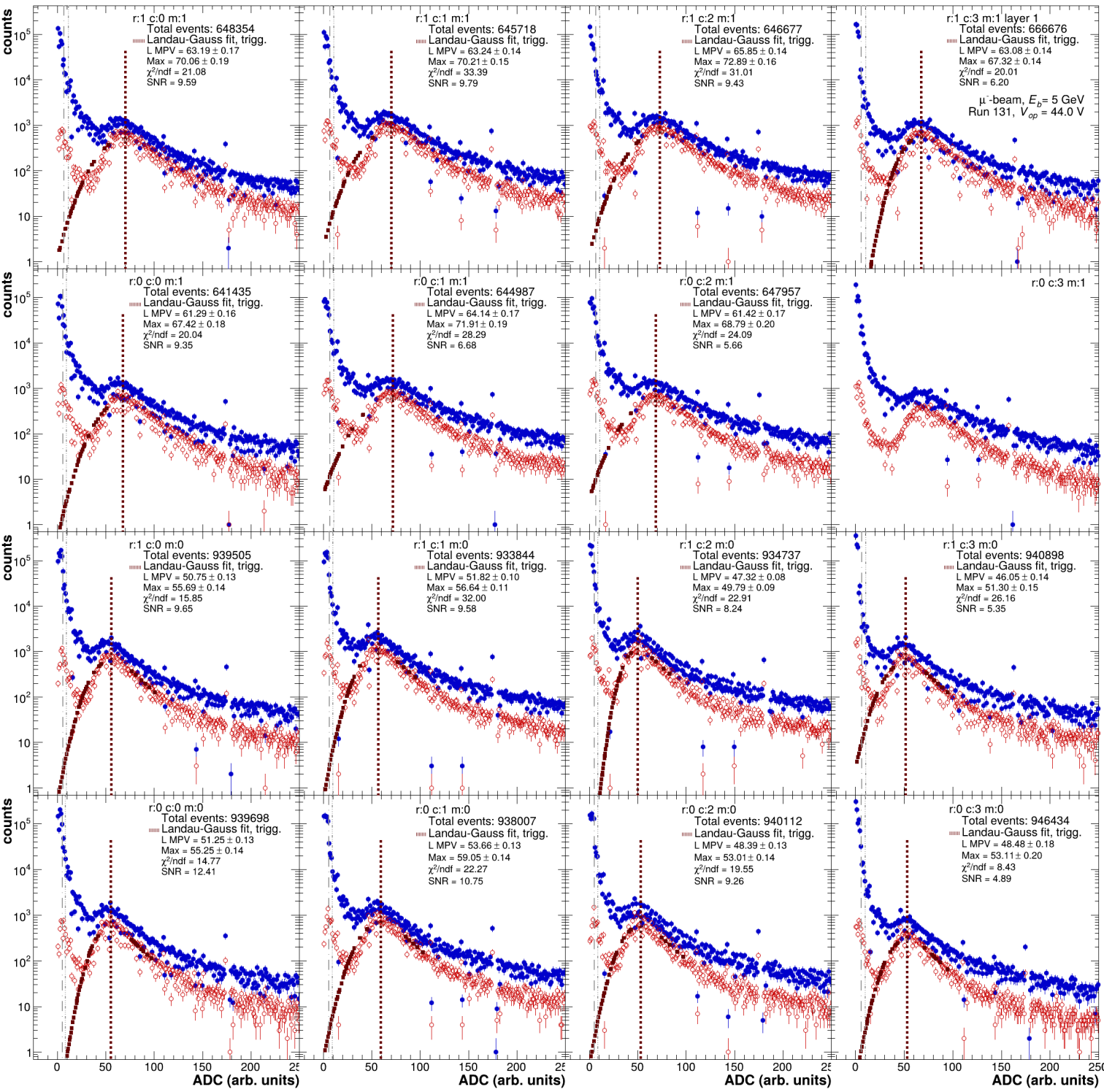

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$MPV_{HG,\,2^{nd}}$ (arb. units)
50
60
70
80
90
100
110
120
130

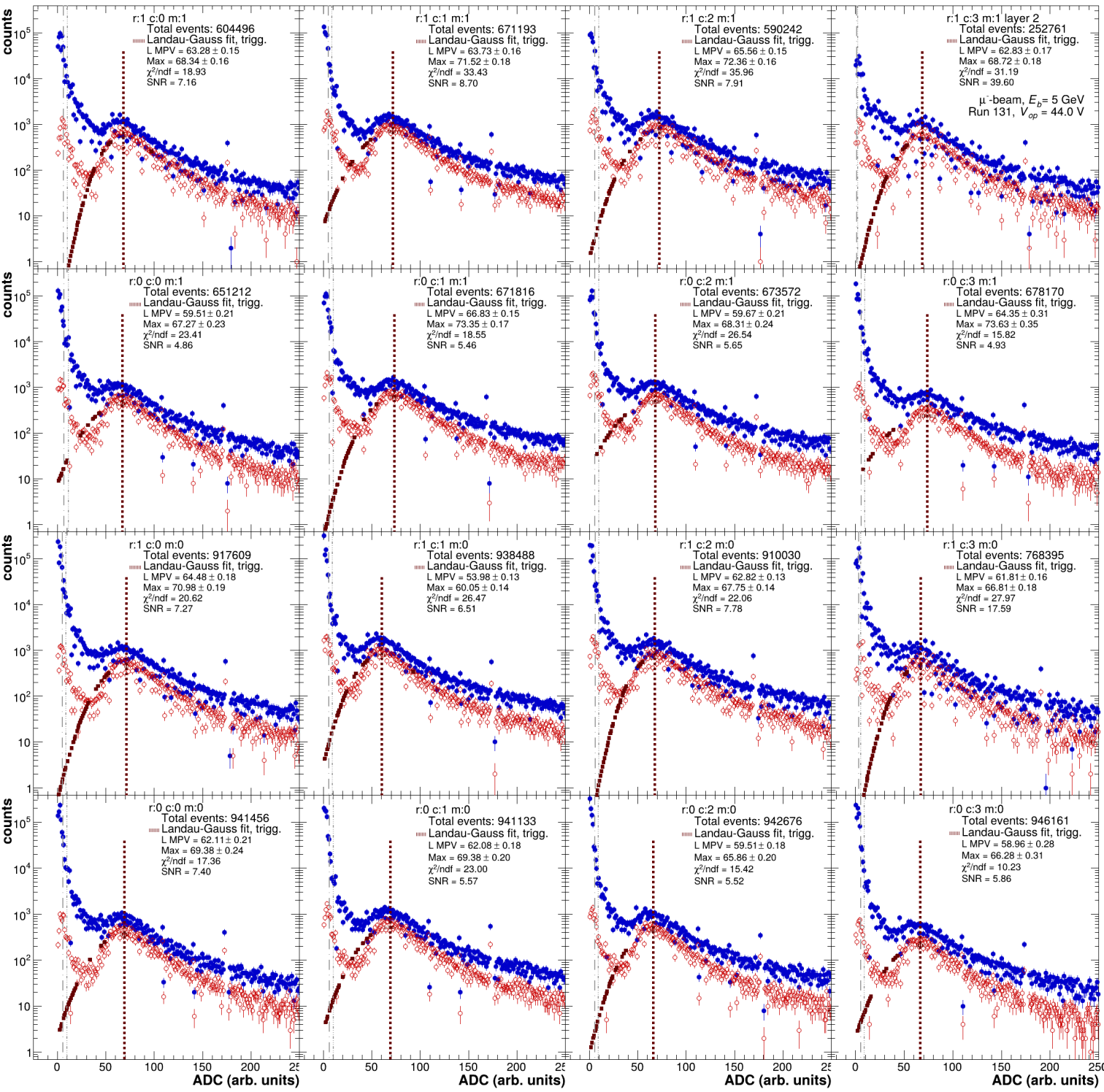

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
3
4
5
6
7
8
9

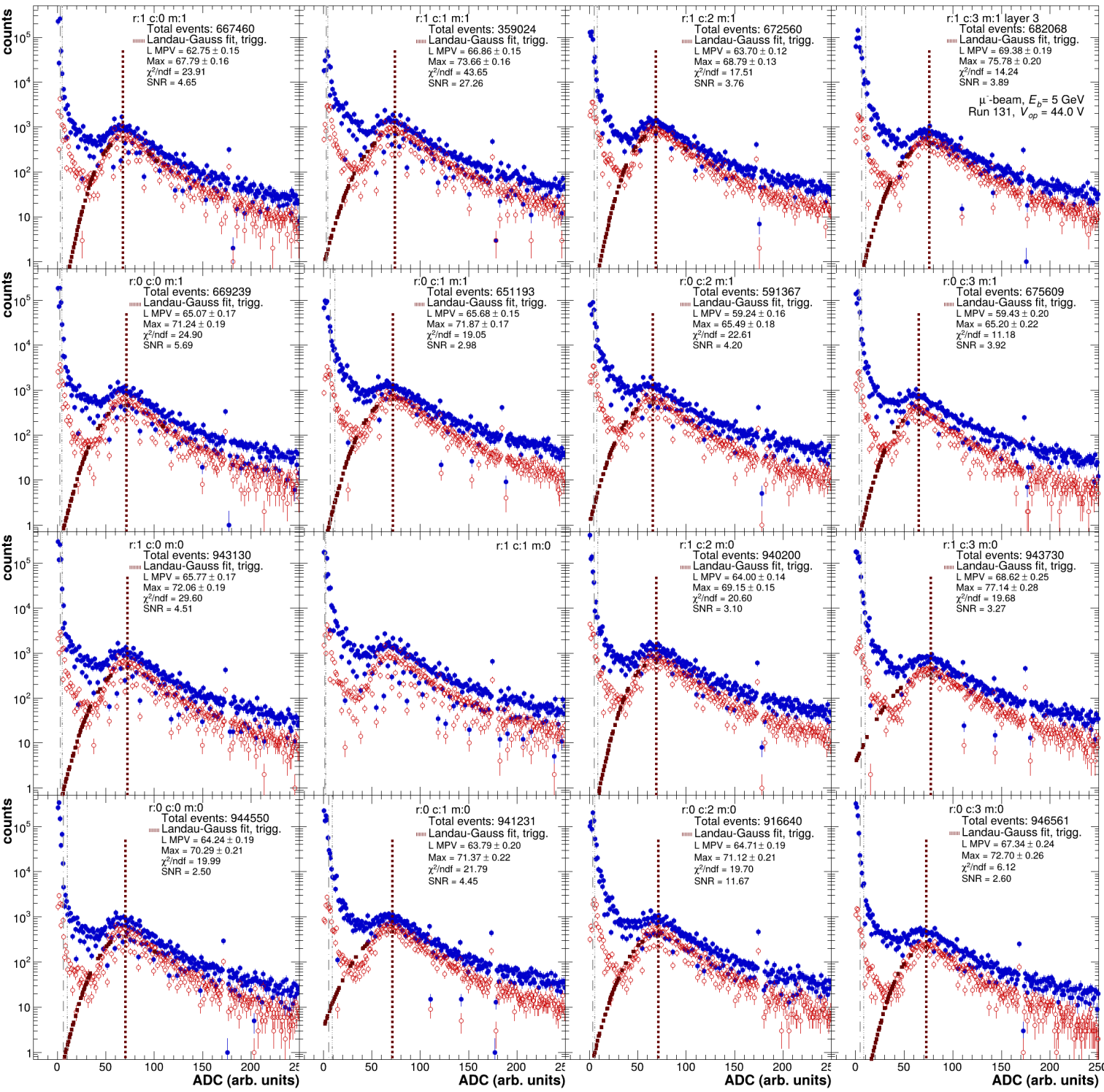

⟨ Max_HG ⟩ = 88.3
μ⁻-beam, E_b = 5 GeV, Run 131, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
50
60
70
80
90
100
110
120
130
140

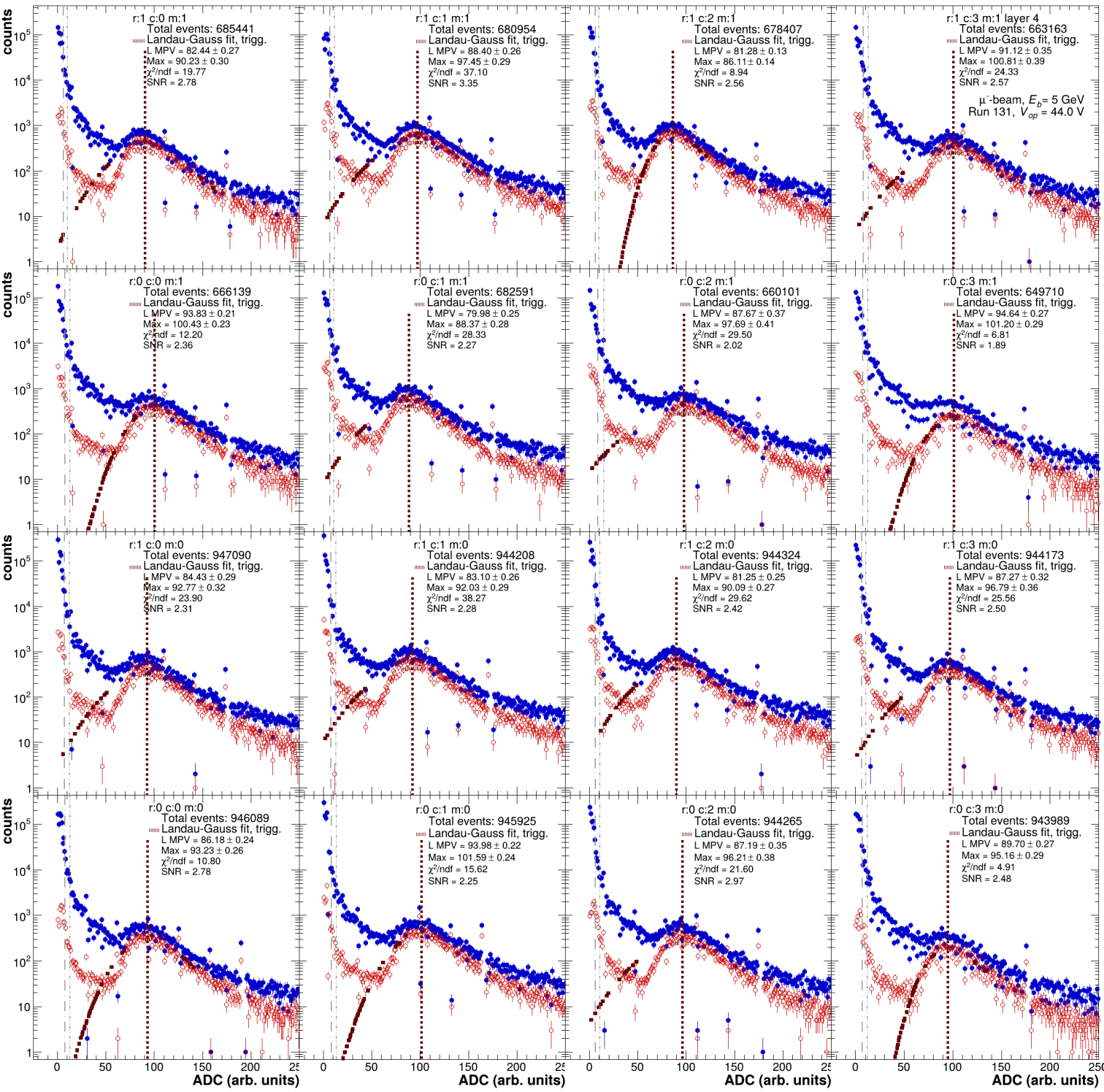

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35
40

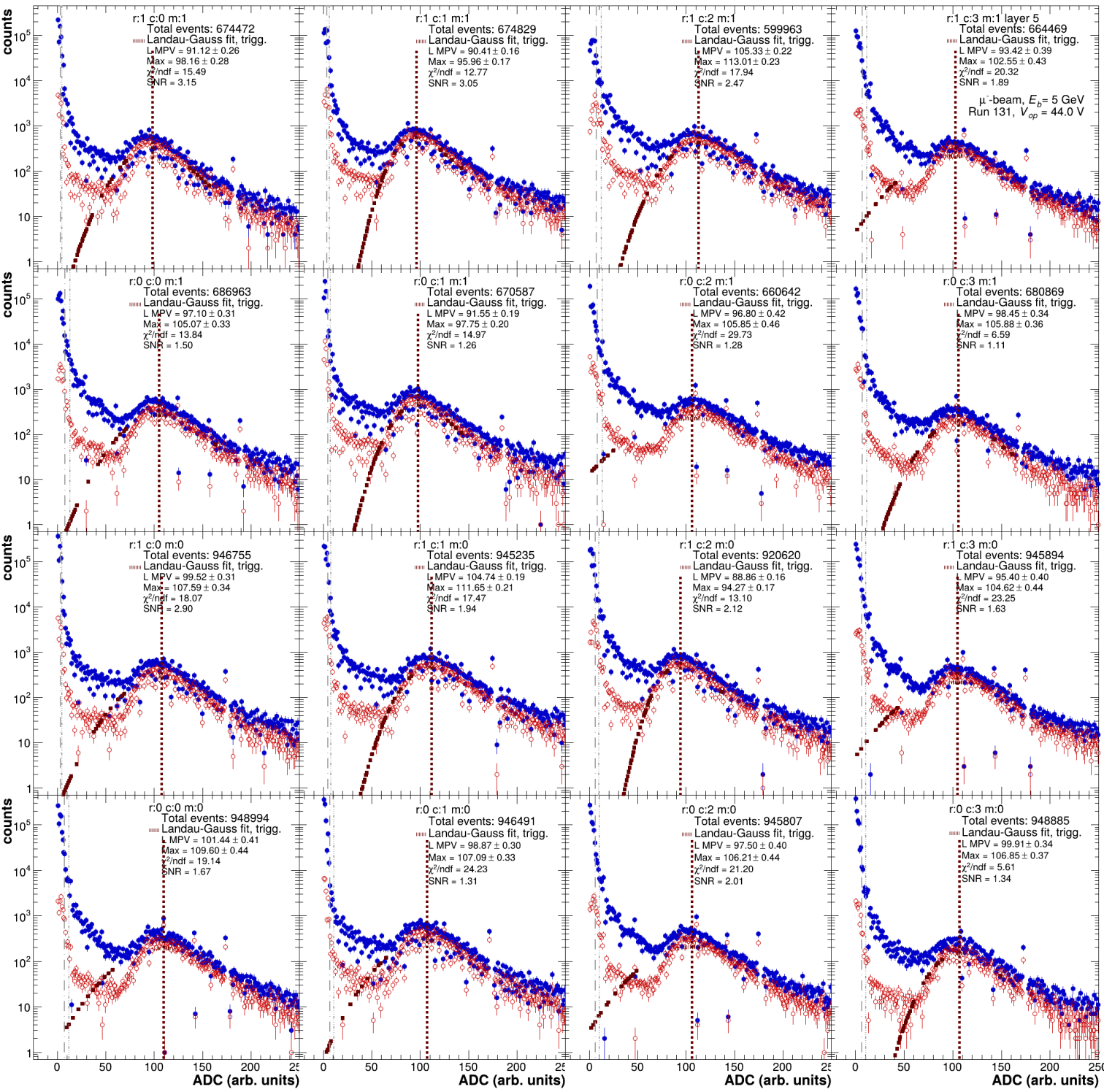

r:1 c:0 m:1
Total events: 651948
Landau-Gauss fit, trigg.
L MPV = 63.62 ± 0.16
Max = 69.76 ± 0.17
χ²/ndf = 18.73
SNR = 7.47
r:1 c:1 m:1
Total events: 598637
Landau-Gauss fit, trigg.
L MPV = 62.35 ± 0.11
Max = 67.40 ± 0.12
χ²/ndf = 31.93
SNR = 10.64
r:1 c:2 m:1
Total events: 542843
Landau-Gauss fit, trigg.
L MPV = 62.79 ± 0.11
Max = 68.18 ± 0.12
χ²/ndf = 43.01
SNR = 14.48
r:1 c:3 m:1 layer 0
Total events: 657910
Landau-Gauss fit, trigg.
L MPV = 64.45 ± 0.17
Max = 71.00 ± 0.19
χ²/ndf = 22.81
SNR = 6.01
μ⁻-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 649121
Landau-Gauss fit, trigg.
L MPV = 64.60 ± 0.15
Max = 69.94 ± 0.16
χ²/ndf = 13.29
SNR = 4.20
r:0 c:1 m:1
Total events: 645304
Landau-Gauss fit, trigg.
L MPV = 58.46 ± 0.12
Max = 63.18 ± 0.12
χ²/ndf = 18.06
SNR = 4.03
r:0 c:2 m:1
Total events: 594400
Landau-Gauss fit, trigg.
L MPV = 59.55 ± 0.15
Max = 65.77 ± 0.17
χ²/ndf = 24.73
SNR = 3.68
r:0 c:3 m:1
Total events: 678066
Landau-Gauss fit, trigg.
L MPV = 52.27 ± 0.19
Max = 57.91 ± 0.21
χ²/ndf = 10.71
SNR = 2.53
r:1 c:0 m:0
Total events: 941797
Landau-Gauss fit, trigg.
L MPV = 45.23 ± 0.11
Max = 49.67 ± 0.12
χ²/ndf = 19.84
SNR = 5.09
r:1 c:1 m:0
Total events: 914832
Landau-Gauss fit, trigg.
L MPV = 49.46 ± 0.08
Max = 52.81 ± 0.09
χ²/ndf = 23.37
SNR = 6.01
r:1 c:2 m:0
Total events: 895514
Landau-Gauss fit, trigg.
L MPV = 47.26 ± 0.08
Max = 50.53 ± 0.09
χ²/ndf = 28.84
SNR = 29.96
r:1 c:3 m:0
Total events: 941532
Landau-Gauss fit, trigg.
L MPV = 46.46 ± 0.11
Max = 50.51 ± 0.12
χ²/ndf = 18.76
SNR = 6.11
r:0 c:0 m:0
Total events: 942354
Landau-Gauss fit, trigg.
L MPV = 49.45 ± 0.13
Max = 53.91 ± 0.14
χ²/ndf = 13.39
SNR = 7.77
r:0 c:1 m:0
Total events: 939560
Landau-Gauss fit, trigg.
L MPV = 49.34 ± 0.10
Max = 53.32 ± 0.11
χ²/ndf = 18.52
SNR = 8.19
r:0 c:2 m:0
Total events: 918003
Landau-Gauss fit, trigg.
L MPV = 47.47 ± 0.11
Max = 51.03 ± 0.12
χ²/ndf = 18.86
SNR = 10.84
r:0 c:3 m:0
Total events: 947515
Landau-Gauss fit, trigg.
L MPV = 43.79 ± 0.16
Max = 47.77 ± 0.17
χ²/ndf = 8.05
SNR = 3.36
counts
ADC (arb. units)

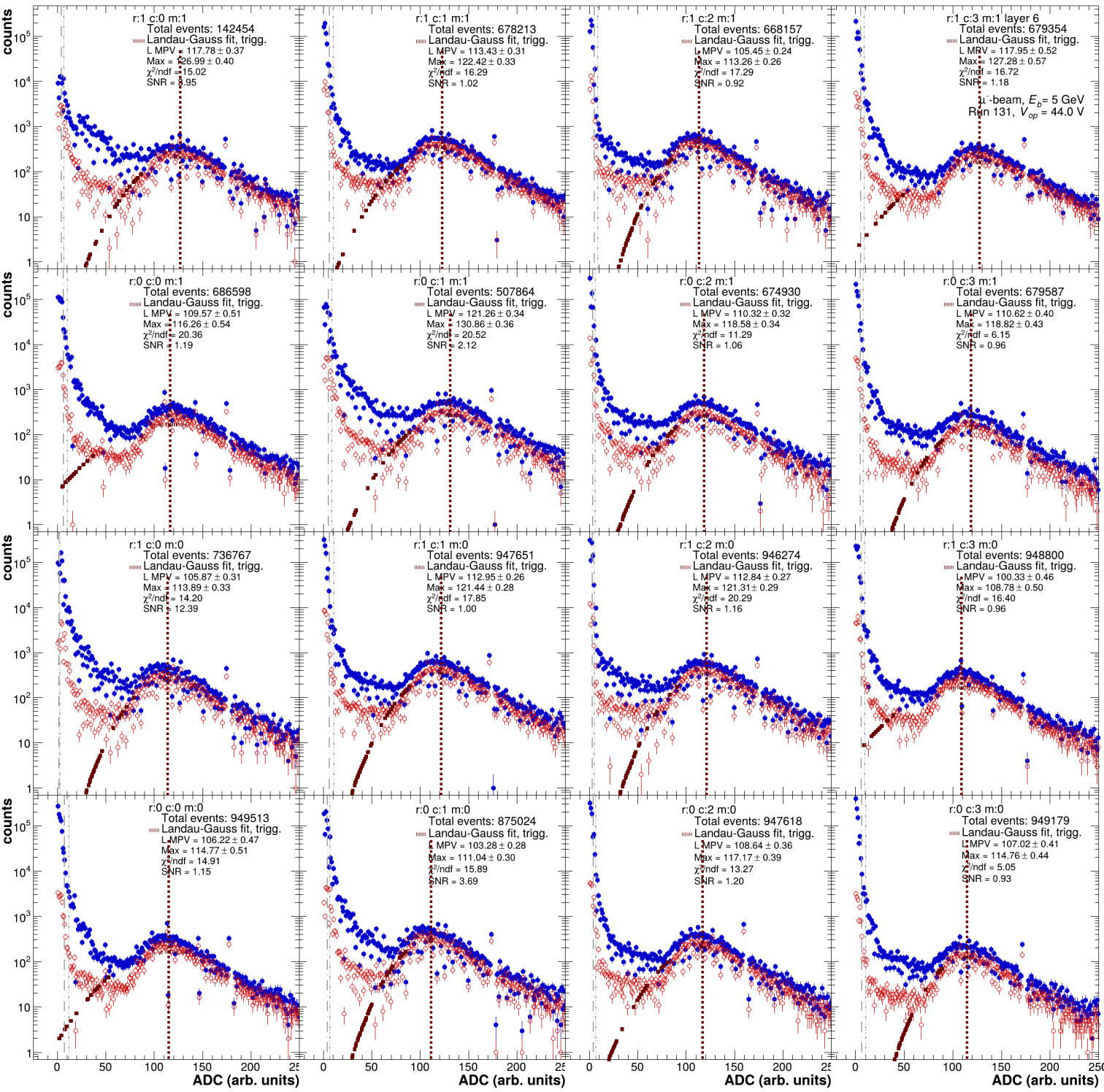

r:1 c:0 m:1
Total events: 648354
Landau-Gauss fit, trigg.
L MPV = 63.19 ± 0.17
Max = 70.06 ± 0.19
χ²/ndf = 21.08
SNR = 9.59
r:1 c:1 m:1
Total events: 645718
Landau-Gauss fit, trigg.
L MPV = 63.24 ± 0.14
Max = 70.21 ± 0.15
χ²/ndf = 33.39
SNR = 9.79
r:1 c:2 m:1
Total events: 646677
Landau-Gauss fit, trigg.
L MPV = 65.85 ± 0.14
Max = 72.89 ± 0.16
χ²/ndf = 31.01
SNR = 9.43
r:1 c:3 m:1 layer 1
Total events: 666676
Landau-Gauss fit, trigg.
L MPV = 63.08 ± 0.14
Max = 67.32 ± 0.14
χ²/ndf = 20.01
SNR = 6.20
μ⁻-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 641435
Landau-Gauss fit, trigg.
L MPV = 61.29 ± 0.16
Max = 67.42 ± 0.18
χ²/ndf = 20.04
SNR = 9.35
r:0 c:1 m:1
Total events: 644987
Landau-Gauss fit, trigg.
L MPV = 64.14 ± 0.17
Max = 71.91 ± 0.19
χ²/ndf = 28.29
SNR = 6.68
r:0 c:2 m:1
Total events: 647957
Landau-Gauss fit, trigg.
L MPV = 61.42 ± 0.17
Max = 68.79 ± 0.20
χ²/ndf = 24.09
SNR = 5.66
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 939505
Landau-Gauss fit, trigg.
L MPV = 50.75 ± 0.13
Max = 55.69 ± 0.14
χ²/ndf = 15.85
SNR = 9.65
r:1 c:1 m:0
Total events: 933844
Landau-Gauss fit, trigg.
L MPV = 51.82 ± 0.10
Max = 56.64 ± 0.11
χ²/ndf = 32.00
SNR = 9.58
r:1 c:2 m:0
Total events: 934737
Landau-Gauss fit, trigg.
L MPV = 47.32 ± 0.08
Max = 49.79 ± 0.09
χ²/ndf = 22.91
SNR = 8.24
r:1 c:3 m:0
Total events: 940898
Landau-Gauss fit, trigg.
L MPV = 46.05 ± 0.14
Max = 51.30 ± 0.15
χ²/ndf = 26.16
SNR = 5.35
r:0 c:0 m:0
Total events: 939698
Landau-Gauss fit, trigg.
L MPV = 51.25 ± 0.13
Max = 55.25 ± 0.14
χ²/ndf = 14.77
SNR = 12.41
r:0 c:1 m:0
Total events: 938007
Landau-Gauss fit, trigg.
L MPV = 53.66 ± 0.13
Max = 59.05 ± 0.14
χ²/ndf = 22.27
SNR = 10.75
r:0 c:2 m:0
Total events: 940112
Landau-Gauss fit, trigg.
L MPV = 48.39 ± 0.13
Max = 53.01 ± 0.14
χ²/ndf = 19.55
SNR = 9.26
r:0 c:3 m:0
Total events: 946434
Landau-Gauss fit, trigg.
L MPV = 48.48 ± 0.18
Max = 53.11 ± 0.20
χ²/ndf = 8.43
SNR = 4.89
counts
ADC (arb. units)

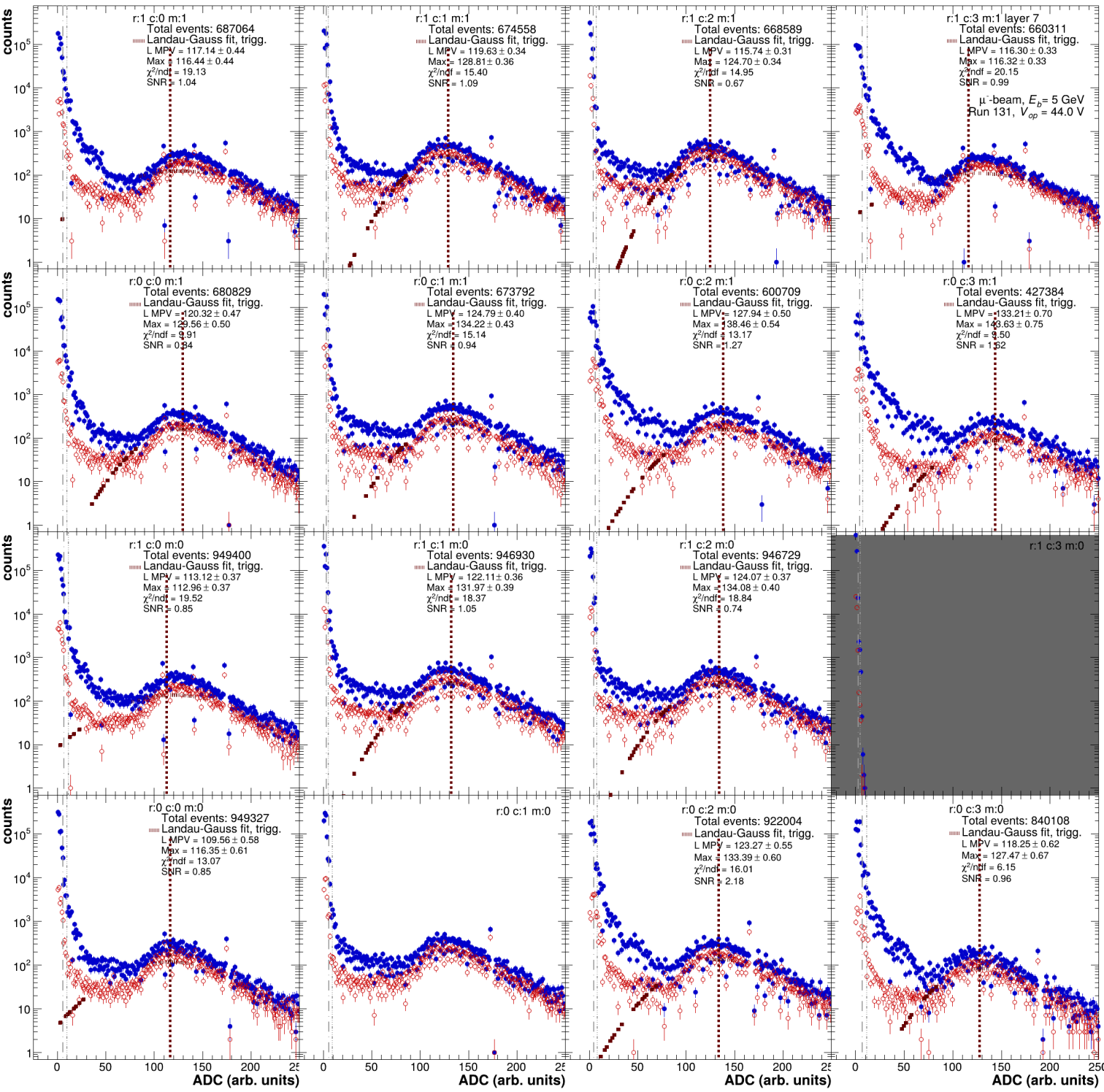

r:1 c:0 m:1
Total events: 604496
Landau-Gauss fit, trigg.
L MPV = 63.28 ± 0.15
Max = 68.34 ± 0.16
χ²/ndf = 18.93
SNR = 7.16
r:1 c:1 m:1
Total events: 671193
Landau-Gauss fit, trigg.
L MPV = 63.73 ± 0.16
Max = 71.52 ± 0.18
χ²/ndf = 33.43
SNR = 8.70
r:1 c:2 m:1
Total events: 590242
Landau-Gauss fit, trigg.
L MPV = 65.56 ± 0.15
Max = 72.36 ± 0.16
χ²/ndf = 35.96
SNR = 7.91
r:1 c:3 m:1 layer 2
Total events: 252761
Landau-Gauss fit, trigg.
L MPV = 62.83 ± 0.17
Max = 68.72 ± 0.18
χ²/ndf = 31.19
SNR = 39.60
μ⁻-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 651212
Landau-Gauss fit, trigg.
L MPV = 59.51 ± 0.21
Max = 67.27 ± 0.23
χ²/ndf = 23.41
SNR = 4.86
r:0 c:1 m:1
Total events: 671816
Landau-Gauss fit, trigg.
L MPV = 66.83 ± 0.15
Max = 73.35 ± 0.17
χ²/ndf = 18.55
SNR = 5.46
r:0 c:2 m:1
Total events: 673572
Landau-Gauss fit, trigg.
L MPV = 59.67 ± 0.21
Max = 68.31 ± 0.24
χ²/ndf = 26.54
SNR = 5.65
r:0 c:3 m:1
Total events: 678170
Landau-Gauss fit, trigg.
L MPV = 64.35 ± 0.31
Max = 73.63 ± 0.35
χ²/ndf = 15.82
SNR = 4.93
r:1 c:0 m:0
Total events: 917609
Landau-Gauss fit, trigg.
L MPV = 64.48 ± 0.18
Max = 70.98 ± 0.19
χ²/ndf = 20.62
SNR = 7.27
r:1 c:1 m:0
Total events: 938488
Landau-Gauss fit, trigg.
L MPV = 53.98 ± 0.13
Max = 60.05 ± 0.14
χ²/ndf = 26.47
SNR = 6.51
r:1 c:2 m:0
Total events: 910030
Landau-Gauss fit, trigg.
L MPV = 62.82 ± 0.13
Max = 67.75 ± 0.14
χ²/ndf = 22.06
SNR = 7.78
r:1 c:3 m:0
Total events: 768395
Landau-Gauss fit, trigg.
L MPV = 61.81 ± 0.16
Max = 66.81 ± 0.18
χ²/ndf = 27.97
SNR = 17.59
r:0 c:0 m:0
Total events: 941456
Landau-Gauss fit, trigg.
L MPV = 62.11 ± 0.21
Max = 69.38 ± 0.24
χ²/ndf = 17.36
SNR = 7.40
r:0 c:1 m:0
Total events: 941133
Landau-Gauss fit, trigg.
L MPV = 62.08 ± 0.18
Max = 69.38 ± 0.20
χ²/ndf = 23.00
SNR = 5.57
r:0 c:2 m:0
Total events: 942676
Landau-Gauss fit, trigg.
L MPV = 59.51 ± 0.18
Max = 65.86 ± 0.20
χ²/ndf = 15.42
SNR = 5.52
r:0 c:3 m:0
Total events: 946161
Landau-Gauss fit, trigg.
L MPV = 58.96 ± 0.28
Max = 66.28 ± 0.31
χ²/ndf = 10.23
SNR = 5.86
counts
ADC (arb. units)

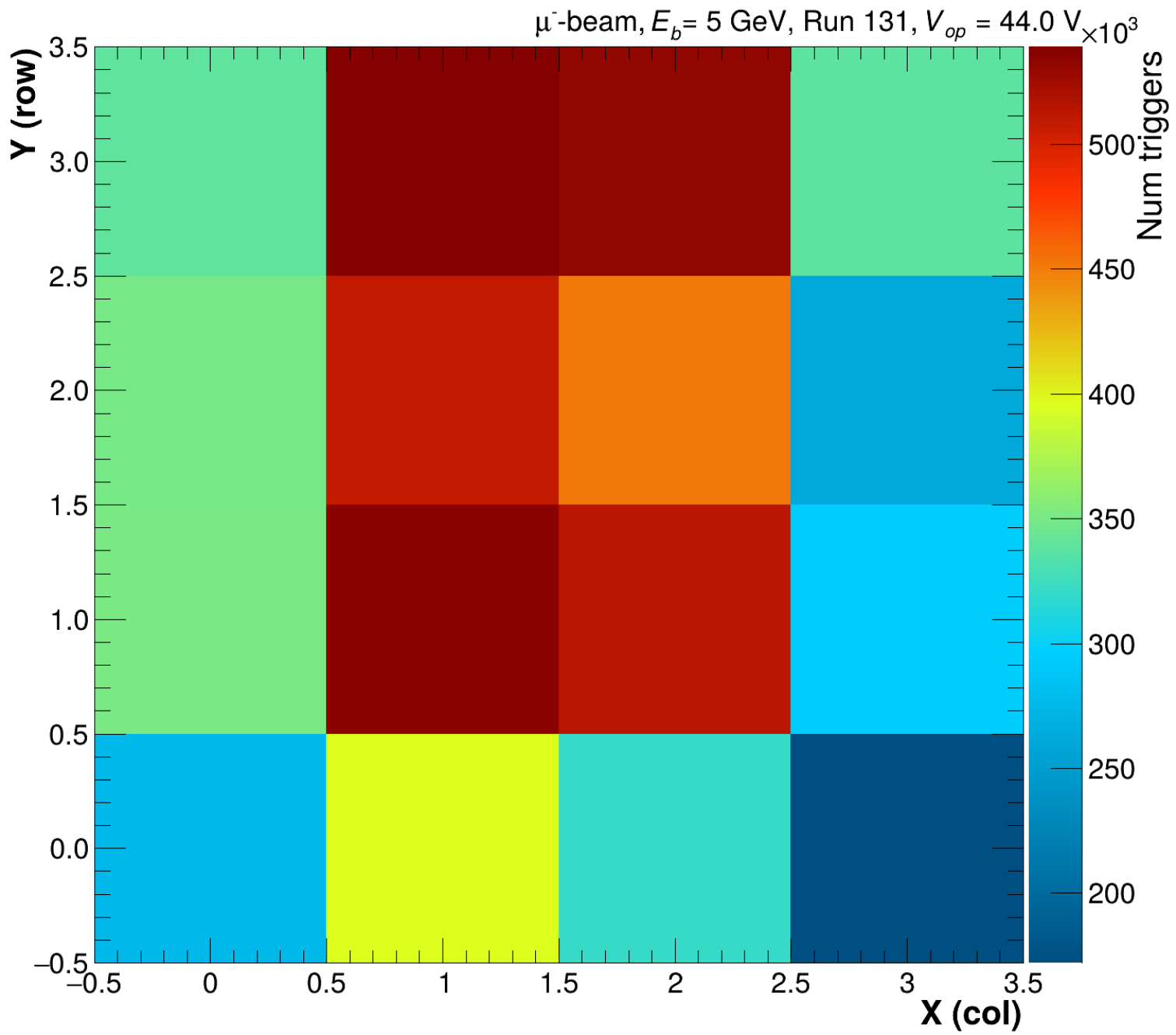

r:1 c:0 m:1
Total events: 667460
Landau-Gauss fit, trigg.
L MPV = 62.75 ± 0.15
Max = 67.79 ± 0.16
χ²/ndf = 23.91
SNR = 4.65
r:1 c:1 m:1
Total events: 359024
Landau-Gauss fit, trigg.
L MPV = 66.86 ± 0.15
Max = 73.66 ± 0.16
χ²/ndf = 43.65
SNR = 27.26
r:1 c:2 m:1
Total events: 672560
Landau-Gauss fit, trigg.
L MPV = 63.70 ± 0.12
Max = 68.79 ± 0.13
χ²/ndf = 17.51
SNR = 3.76
r:1 c:3 m:1 layer 3
Total events: 682068
Landau-Gauss fit, trigg.
L MPV = 69.38 ± 0.19
Max = 75.78 ± 0.20
χ²/ndf = 14.24
SNR = 3.89
μ⁻-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 669239
Landau-Gauss fit, trigg.
L MPV = 65.07 ± 0.17
Max = 71.24 ± 0.19
χ²/ndf = 24.90
SNR = 5.69
r:0 c:1 m:1
Total events: 651193
Landau-Gauss fit, trigg.
L MPV = 65.68 ± 0.15
Max = 71.87 ± 0.17
χ²/ndf = 19.05
SNR = 2.98
r:0 c:2 m:1
Total events: 591367
Landau-Gauss fit, trigg.
L MPV = 59.24 ± 0.16
Max = 65.49 ± 0.18
χ²/ndf = 22.61
SNR = 4.20
r:0 c:3 m:1
Total events: 675609
Landau-Gauss fit, trigg.
L MPV = 59.43 ± 0.20
Max = 65.20 ± 0.22
χ²/ndf = 11.18
SNR = 3.92
r:1 c:0 m:0
Total events: 943130
Landau-Gauss fit, trigg.
L MPV = 65.77 ± 0.17
Max = 72.06 ± 0.19
χ²/ndf = 29.60
SNR = 4.51
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 940200
Landau-Gauss fit, trigg.
L MPV = 64.00 ± 0.14
Max = 69.15 ± 0.15
χ²/ndf = 20.60
SNR = 3.10
r:1 c:3 m:0
Total events: 943730
Landau-Gauss fit, trigg.
L MPV = 68.62 ± 0.25
Max = 77.14 ± 0.28
χ²/ndf = 19.68
SNR = 3.27
r:0 c:0 m:0
Total events: 944550
Landau-Gauss fit, trigg.
L MPV = 64.24 ± 0.19
Max = 70.29 ± 0.21
χ²/ndf = 19.99
SNR = 2.50
r:0 c:1 m:0
Total events: 941231
Landau-Gauss fit, trigg.
L MPV = 63.79 ± 0.20
Max = 71.37 ± 0.22
χ²/ndf = 21.79
SNR = 4.45
r:0 c:2 m:0
Total events: 916640
Landau-Gauss fit, trigg.
L MPV = 64.71 ± 0.19
Max = 71.12 ± 0.21
χ²/ndf = 19.70
SNR = 11.67
r:0 c:3 m:0
Total events: 946561
Landau-Gauss fit, trigg.
L MPV = 67.34 ± 0.24
Max = 72.70 ± 0.26
χ²/ndf = 6.12
SNR = 2.60
counts
ADC (arb. units)

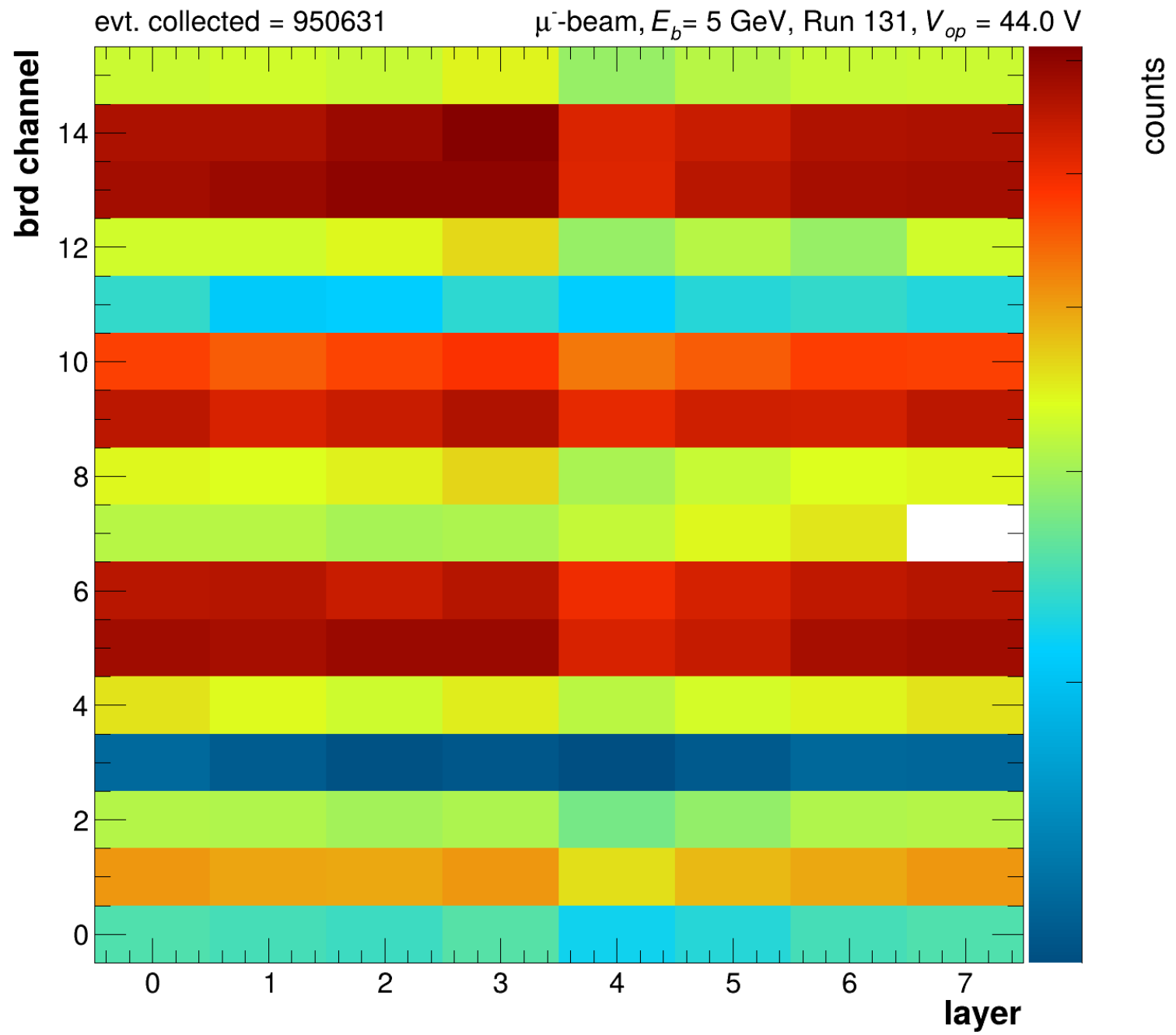

r:1 c:0 m:1
Total events: 685441
Landau-Gauss fit, trigg.
L MPV = 82.44 ± 0.27
Max = 90.23 ± 0.30
χ²/ndf = 19.77
SNR = 2.78
r:1 c:1 m:1
Total events: 680954
Landau-Gauss fit, trigg.
L MPV = 88.40 ± 0.26
Max = 97.45 ± 0.29
χ²/ndf = 37.10
SNR = 3.35
r:1 c:2 m:1
Total events: 678407
Landau-Gauss fit, trigg.
L MPV = 81.28 ± 0.13
Max = 86.11 ± 0.14
χ²/ndf = 8.94
SNR = 2.56
r:1 c:3 m:1 layer 4
Total events: 663163
Landau-Gauss fit, trigg.
L MPV = 91.12 ± 0.35
Max = 100.81 ± 0.39
χ²/ndf = 24.33
SNR = 2.57
μ-beam, E_b = 5 GeV
Run 131, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 666139
Landau-Gauss fit, trigg.
L MPV = 93.83 ± 0.21
Max = 100.43 ± 0.23
χ²/ndf = 12.20
SNR = 2.36
r:0 c:1 m:1
Total events: 682591
Landau-Gauss fit, trigg.
L MPV = 79.98 ± 0.25
Max = 88.37 ± 0.28
χ²/ndf = 28.33
SNR = 2.27
r:0 c:2 m:1
Total events: 660101
Landau-Gauss fit, trigg.
L MPV = 87.67 ± 0.37
Max = 97.69 ± 0.41
χ²/ndf = 29.50
SNR = 2.02
r:0 c:3 m:1
Total events: 649710
Landau-Gauss fit, trigg.
L MPV = 94.64 ± 0.27
Max = 101.20 ± 0.29
χ²/ndf = 6.81
SNR = 1.89
r:1 c:0 m:0
Total events: 947090
Landau-Gauss fit, trigg.
L MPV = 84.43 ± 0.29
Max = 92.77 ± 0.32
χ²/ndf = 23.90
SNR = 2.31
r:1 c:1 m:0
Total events: 944208
Landau-Gauss fit, trigg.
L MPV = 83.10 ± 0.26
Max = 92.03 ± 0.29
χ²/ndf = 38.27
SNR = 2.28
r:1 c:2 m:0
Total events: 944324
Landau-Gauss fit, trigg.
L MPV = 81.25 ± 0.25
Max = 90.09 ± 0.27
χ²/ndf = 29.62
SNR = 2.42
r:1 c:3 m:0
Total events: 944173
Landau-Gauss fit, trigg.
L MPV = 87.27 ± 0.32
Max = 96.79 ± 0.36
χ²/ndf = 25.56
SNR = 2.50
r:0 c:0 m:0
Total events: 946089
Landau-Gauss fit, trigg.
L MPV = 86.18 ± 0.24
Max = 93.23 ± 0.26
χ²/ndf = 10.80
SNR = 2.78
r:0 c:1 m:0
Total events: 945925
Landau-Gauss fit, trigg.
L MPV = 93.98 ± 0.22
Max = 101.59 ± 0.24
χ²/ndf = 15.62
SNR = 2.25
r:0 c:2 m:0
Total events: 944265
Landau-Gauss fit, trigg.
L MPV = 87.19 ± 0.35
Max = 96.21 ± 0.38
χ²/ndf = 21.60
SNR = 2.97
r:0 c:3 m:0
Total events: 943989
Landau-Gauss fit, trigg.
L MPV = 89.70 ± 0.27
Max = 95.16 ± 0.29
χ²/ndf = 4.91
SNR = 2.48
counts
ADC (arb. units)

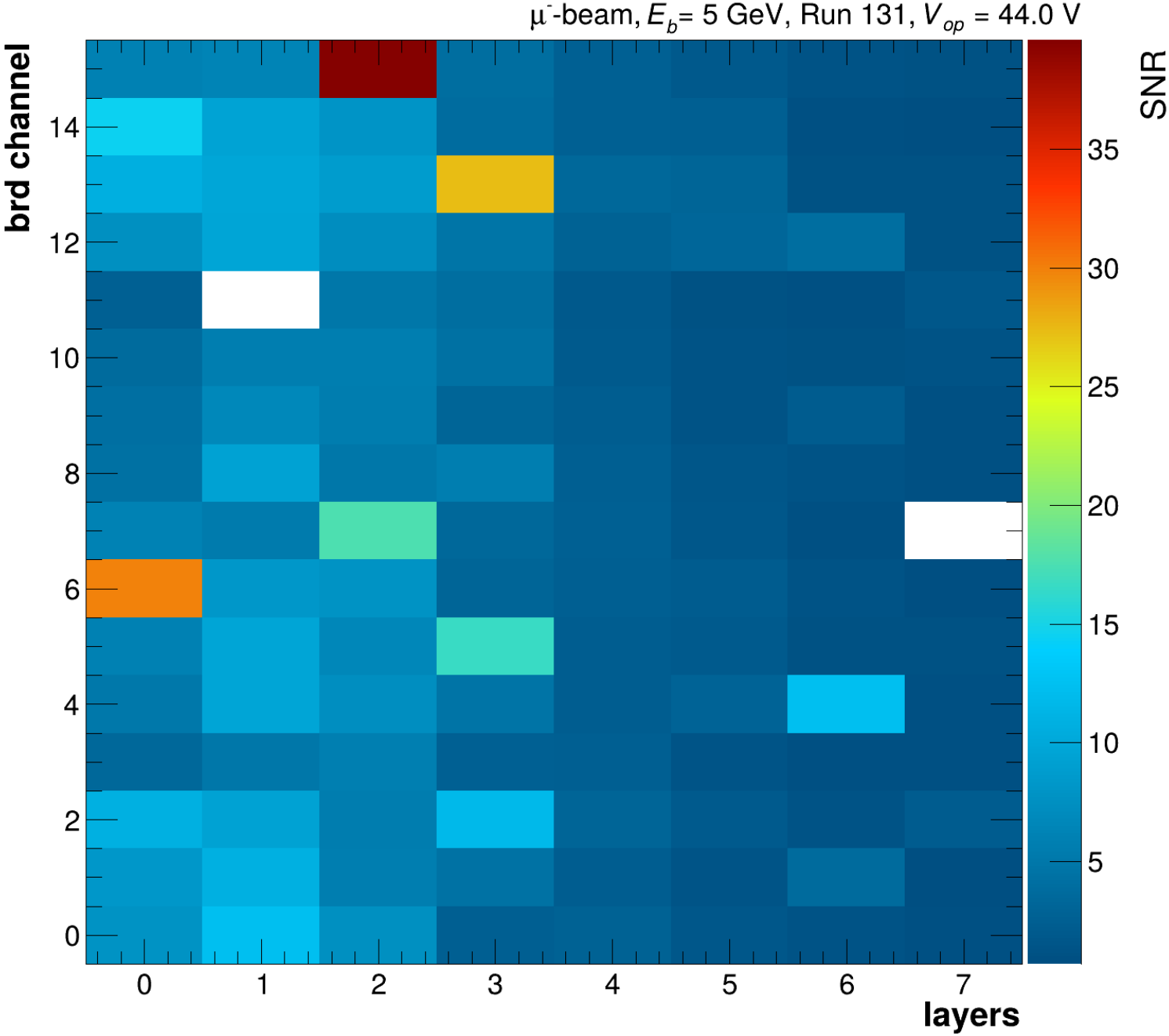

r:1 c:0 m:1
Total events: 674472
Landau-Gauss fit, trigg.
L MPV = 91.12 ± 0.26
Max = 98.16 ± 0.28
χ²/ndf = 15.49
SNR = 3.15
r:1 c:1 m:1
Total events: 674829
Landau-Gauss fit, trigg.
L MPV = 90.41 ± 0.16
Max = 95.96 ± 0.17
χ²/ndf = 12.77
SNR = 3.05
r:1 c:2 m:1
Total events: 599963
Landau-Gauss fit, trigg.
L MPV = 105.33 ± 0.22
Max = 113.01 ± 0.23
χ²/ndf = 17.94
SNR = 2.47
r:1 c:3 m:1 layer 5
Total events: 664469
Landau-Gauss fit, trigg.
L MPV = 93.42 ± 0.39
Max = 102.55 ± 0.43
χ²/ndf = 20.32
SNR = 1.89
μ⁻-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 686963
Landau-Gauss fit, trigg.
L MPV = 97.10 ± 0.31
Max = 105.07 ± 0.33
χ²/ndf = 13.84
SNR = 1.50
r:0 c:1 m:1
Total events: 670587
Landau-Gauss fit, trigg.
L MPV = 91.55 ± 0.19
Max = 97.75 ± 0.20
χ²/ndf = 14.97
SNR = 1.26
r:0 c:2 m:1
Total events: 660642
Landau-Gauss fit, trigg.
L MPV = 96.80 ± 0.42
Max = 105.85 ± 0.46
χ²/ndf = 29.73
SNR = 1.28
r:0 c:3 m:1
Total events: 680869
Landau-Gauss fit, trigg.
L MPV = 98.45 ± 0.34
Max = 105.88 ± 0.36
χ²/ndf = 6.59
SNR = 1.11
r:1 c:0 m:0
Total events: 946755
Landau-Gauss fit, trigg.
L MPV = 99.52 ± 0.31
Max = 107.59 ± 0.34
χ²/ndf = 18.07
SNR = 2.90
r:1 c:1 m:0
Total events: 945235
Landau-Gauss fit, trigg.
L MPV = 104.74 ± 0.19
Max = 111.65 ± 0.21
χ²/ndf = 17.47
SNR = 1.94
r:1 c:2 m:0
Total events: 920620
Landau-Gauss fit, trigg.
L MPV = 88.86 ± 0.16
Max = 94.27 ± 0.17
χ²/ndf = 13.10
SNR = 2.12
r:1 c:3 m:0
Total events: 945894
Landau-Gauss fit, trigg.
L MPV = 95.40 ± 0.40
Max = 104.62 ± 0.44
χ²/ndf = 23.25
SNR = 1.63
r:0 c:0 m:0
Total events: 948994
Landau-Gauss fit, trigg.
L MPV = 101.44 ± 0.41
Max = 109.60 ± 0.44
χ²/ndf = 19.14
SNR = 1.67
r:0 c:1 m:0
Total events: 946491
Landau-Gauss fit, trigg.
L MPV = 98.87 ± 0.30
Max = 107.09 ± 0.33
χ²/ndf = 24.23
SNR = 1.31
r:0 c:2 m:0
Total events: 945807
Landau-Gauss fit, trigg.
L MPV = 97.50 ± 0.40
Max = 106.21 ± 0.44
χ²/ndf = 21.20
SNR = 2.01
r:0 c:3 m:0
Total events: 948885
Landau-Gauss fit, trigg.
L MPV = 99.91 ± 0.34
Max = 106.85 ± 0.37
χ²/ndf = 5.61
SNR = 1.34
counts
ADC (arb. units)

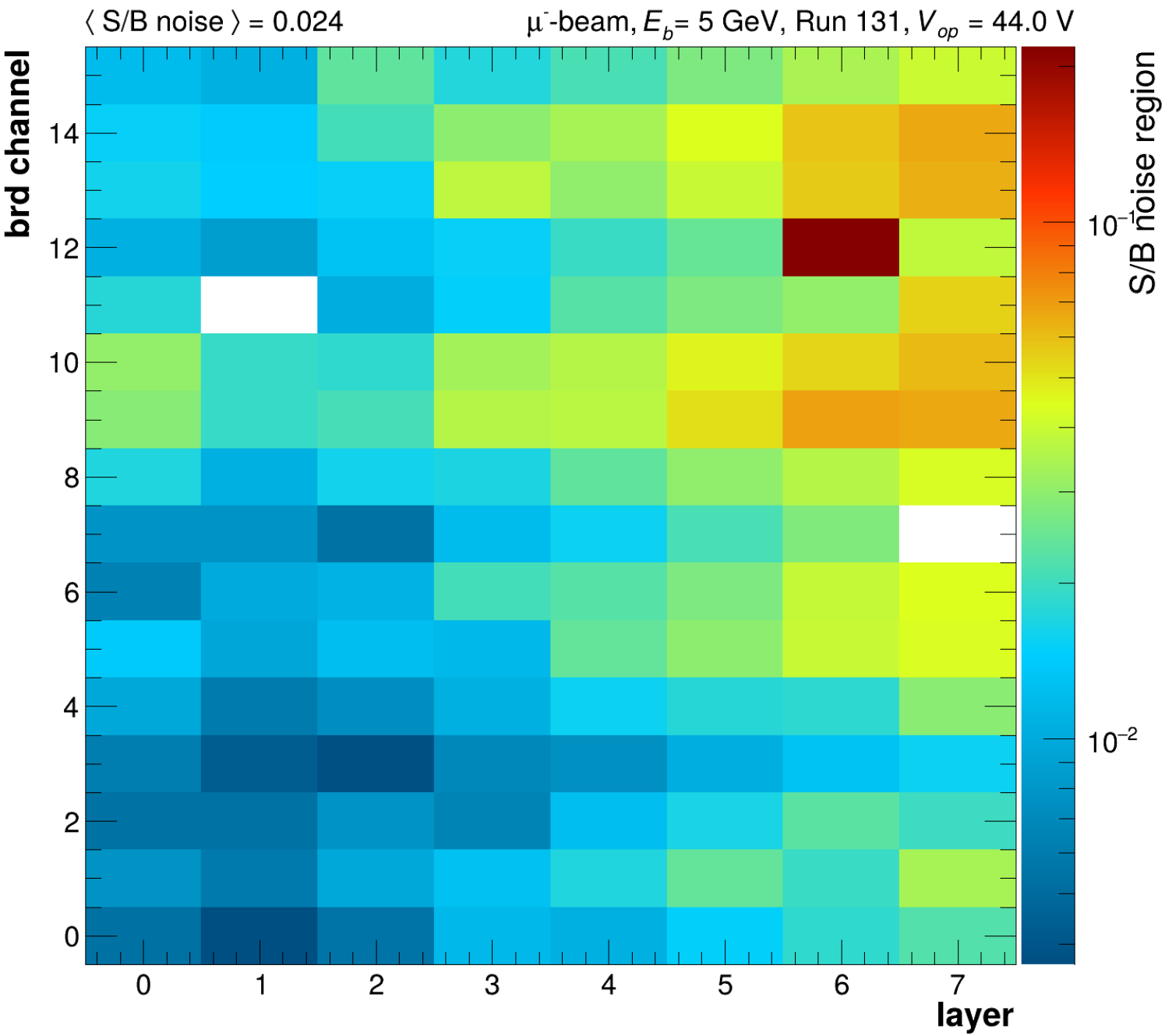

r:1 c:0 m:1
Total events: 142454
Landau-Gauss fit, trigg.
L MPV = 117.78 ± 0.37
Max = 126.99 ± 0.40
χ²/ndf = 15.02
SNR = 3.95
r:1 c:1 m:1
Total events: 678213
Landau-Gauss fit, trigg.
L MPV = 113.43 ± 0.31
Max = 122.42 ± 0.33
χ²/ndf = 16.29
SNR = 1.02
r:1 c:2 m:1
Total events: 668157
Landau-Gauss fit, trigg.
L MPV = 105.45 ± 0.24
Max = 113.26 ± 0.26
χ²/ndf = 17.29
SNR = 0.92
r:1 c:3 m:1 layer 6
Total events: 679354
Landau-Gauss fit, trigg.
L MPV = 117.95 ± 0.52
Max = 127.28 ± 0.57
χ²/ndf = 16.72
SNR = 1.18
μ-beam, E_b = 5 GeV
Run 131, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 686598
Landau-Gauss fit, trigg.
L MPV = 109.57 ± 0.51
Max = 116.26 ± 0.54
χ²/ndf = 20.36
SNR = 1.19
r:0 c:1 m:1
Total events: 507864
Landau-Gauss fit, trigg.
L MPV = 121.26 ± 0.34
Max = 130.86 ± 0.36
χ²/ndf = 20.52
SNR = 2.12
r:0 c:2 m:1
Total events: 674930
Landau-Gauss fit, trigg.
L MPV = 110.32 ± 0.32
Max = 118.58 ± 0.34
χ²/ndf = 11.29
SNR = 1.06
r:0 c:3 m:1
Total events: 679587
Landau-Gauss fit, trigg.
L MPV = 110.62 ± 0.40
Max = 118.82 ± 0.43
χ²/ndf = 6.15
SNR = 0.96
r:1 c:0 m:0
Total events: 736767
Landau-Gauss fit, trigg.
L MPV = 105.87 ± 0.31
Max = 113.89 ± 0.33
χ²/ndf = 14.20
SNR = 12.39
r:1 c:1 m:0
Total events: 947651
Landau-Gauss fit, trigg.
L MPV = 112.95 ± 0.26
Max = 121.44 ± 0.28
χ²/ndf = 17.85
SNR = 1.00
r:1 c:2 m:0
Total events: 946274
Landau-Gauss fit, trigg.
L MPV = 112.84 ± 0.27
Max = 121.31 ± 0.29
χ²/ndf = 20.29
SNR = 1.16
r:1 c:3 m:0
Total events: 948800
Landau-Gauss fit, trigg.
L MPV = 100.33 ± 0.46
Max = 108.78 ± 0.50
χ²/ndf = 16.40
SNR = 0.96
r:0 c:0 m:0
Total events: 949513
Landau-Gauss fit, trigg.
L MPV = 106.22 ± 0.47
Max = 114.77 ± 0.51
χ²/ndf = 14.91
SNR = 1.15
r:0 c:1 m:0
Total events: 875024
Landau-Gauss fit, trigg.
L MPV = 103.28 ± 0.28
Max = 111.04 ± 0.30
χ²/ndf = 15.89
SNR = 3.69
r:0 c:2 m:0
Total events: 947618
Landau-Gauss fit, trigg.
L MPV = 108.64 ± 0.36
Max = 117.17 ± 0.39
χ²/ndf = 13.27
SNR = 1.20
r:0 c:3 m:0
Total events: 949179
Landau-Gauss fit, trigg.
L MPV = 107.02 ± 0.41
Max = 114.76 ± 0.44
χ²/ndf = 5.05
SNR = 0.93
counts
ADC (arb. units)

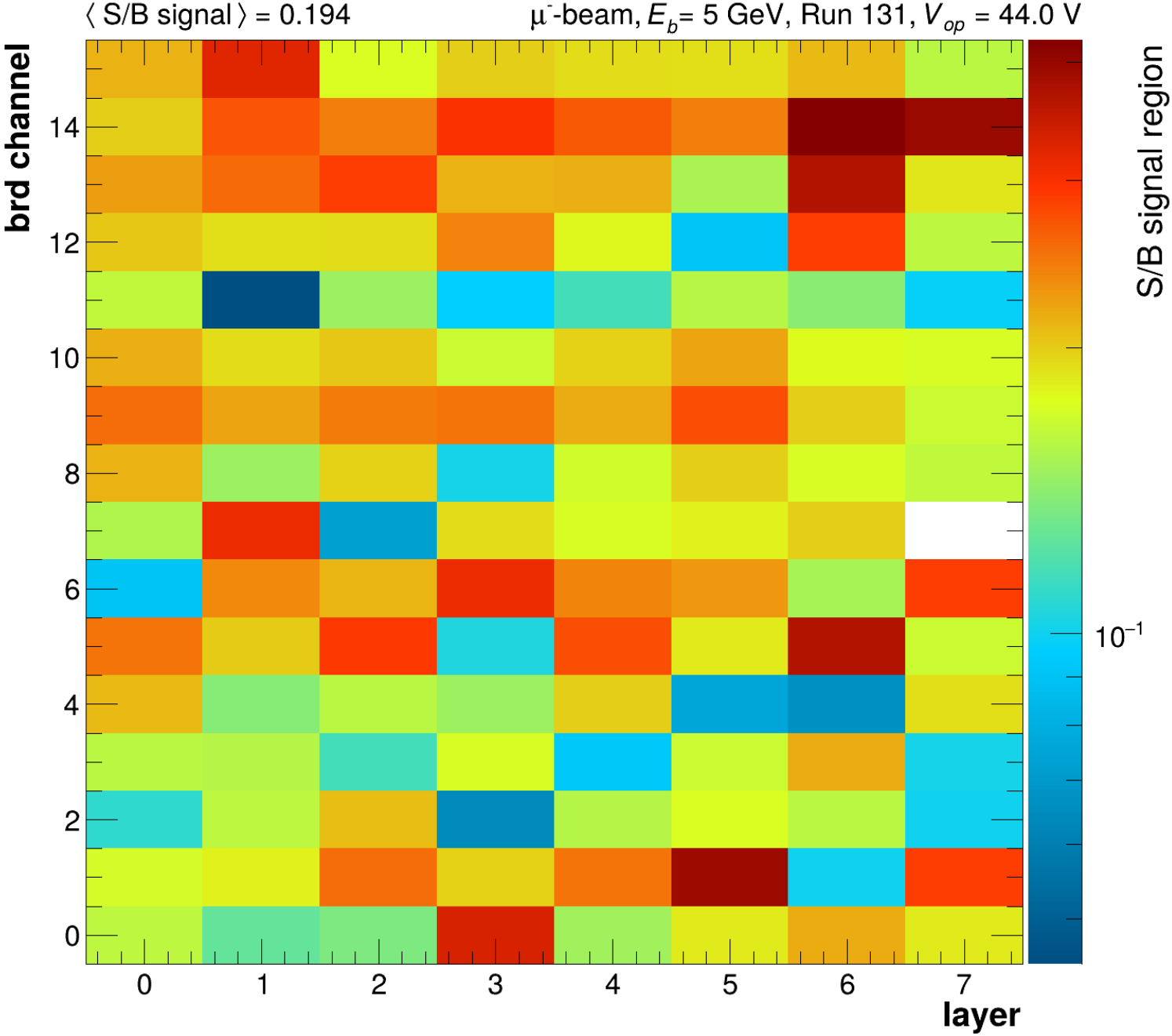

r:1 c:0 m:1
Total events: 687064
Landau-Gauss fit, trigg.
L MPV = 117.14 ± 0.44
Max = 116.44 ± 0.44
χ²/ndf = 19.13
SNR = 1.04
r:1 c:1 m:1
Total events: 674558
Landau-Gauss fit, trigg.
L MPV = 119.63 ± 0.34
Max = 128.81 ± 0.36
χ²/ndf = 15.40
SNR = 1.09
r:1 c:2 m:1
Total events: 668589
Landau-Gauss fit, trigg.
L MPV = 115.74 ± 0.31
Max = 124.70 ± 0.34
χ²/ndf = 14.95
SNR = 0.67
r:1 c:3 m:1 layer 7
Total events: 660311
Landau-Gauss fit, trigg.
L MPV = 116.30 ± 0.33
Max = 116.32 ± 0.33
χ²/ndf = 20.15
SNR = 0.99
μ-beam, E_b = 5 GeV
Run 131, V_op = 44.0 V
r:0 c:0 m:1
Total events: 680829
Landau-Gauss fit, trigg.
L MPV = 120.32 ± 0.47
Max = 129.56 ± 0.50
χ²/ndf = 9.91
SNR = 0.84
r:0 c:1 m:1
Total events: 673792
Landau-Gauss fit, trigg.
L MPV = 124.79 ± 0.40
Max = 134.22 ± 0.43
χ²/ndf = 15.14
SNR = 0.94
r:0 c:2 m:1
Total events: 600709
Landau-Gauss fit, trigg.
L MPV = 127.94 ± 0.50
Max = 138.46 ± 0.54
χ²/ndf = 13.17
SNR = 1.27
r:0 c:3 m:1
Total events: 427384
Landau-Gauss fit, trigg.
L MPV = 133.21 ± 0.70
Max = 143.63 ± 0.75
χ²/ndf = 9.50
SNR = 1.52
r:1 c:0 m:0
Total events: 949400
Landau-Gauss fit, trigg.
L MPV = 113.12 ± 0.37
Max = 112.96 ± 0.37
χ²/ndf = 19.52
SNR = 0.85
r:1 c:1 m:0
Total events: 946930
Landau-Gauss fit, trigg.
L MPV = 122.11 ± 0.36
Max = 131.97 ± 0.39
χ²/ndf = 18.37
SNR = 1.05
r:1 c:2 m:0
Total events: 946729
Landau-Gauss fit, trigg.
L MPV = 124.07 ± 0.37
Max = 134.08 ± 0.40
χ²/ndf = 18.84
SNR = 0.74
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 949327
Landau-Gauss fit, trigg.
L MPV = 109.56 ± 0.58
Max = 116.35 ± 0.61
χ²/ndf = 13.07
SNR = 0.85
r:0 c:1 m:0
r:0 c:2 m:0
Total events: 922004
Landau-Gauss fit, trigg.
L MPV = 123.27 ± 0.55
Max = 133.39 ± 0.60
χ²/ndf = 16.01
SNR = 2.18
r:0 c:3 m:0
Total events: 840108
Landau-Gauss fit, trigg.
L MPV = 118.25 ± 0.62
Max = 127.47 ± 0.67
χ²/ndf = 6.15
SNR = 0.96
counts
ADC (arb. units)

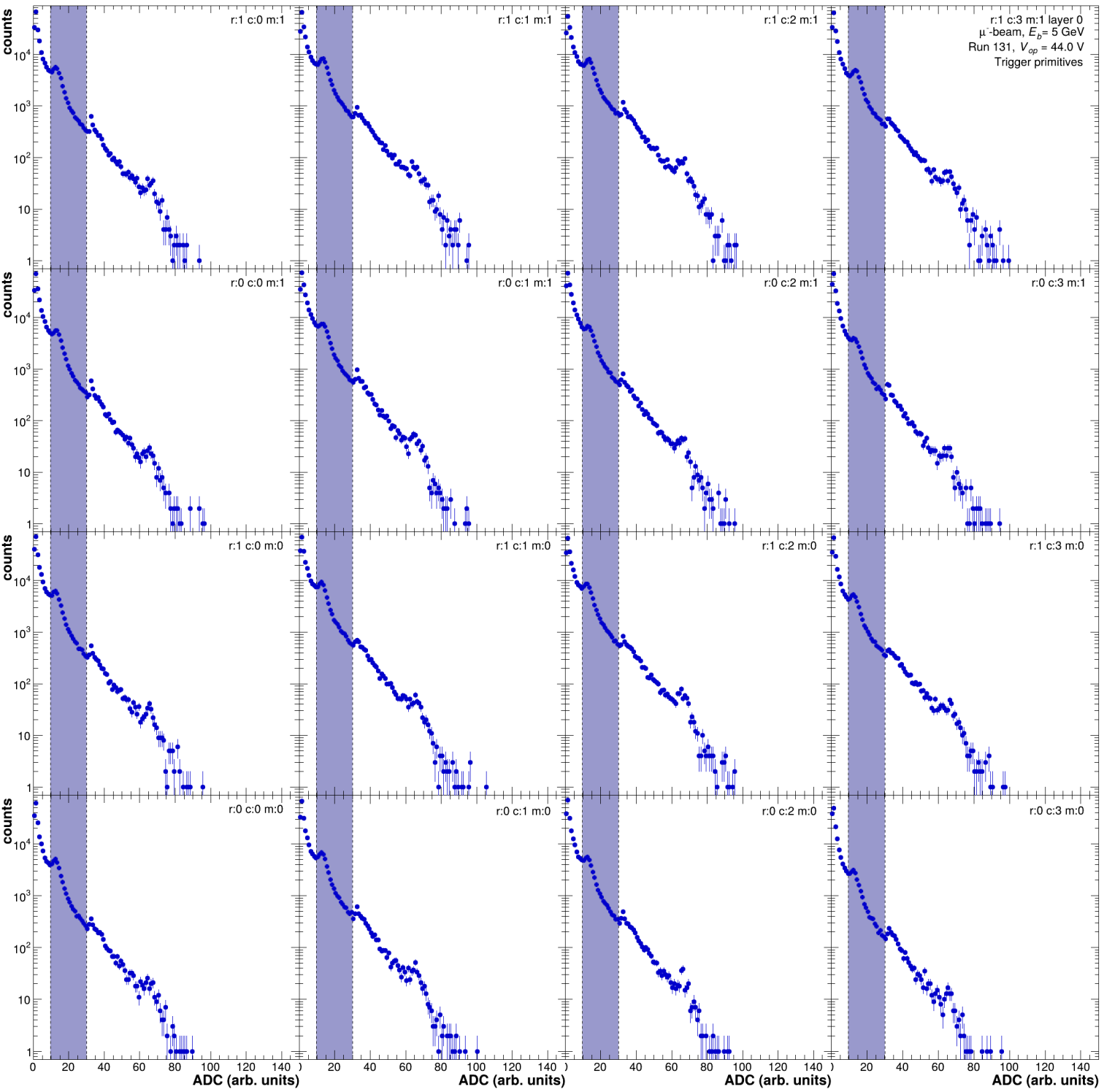

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
500
450
400
350
300
250
200

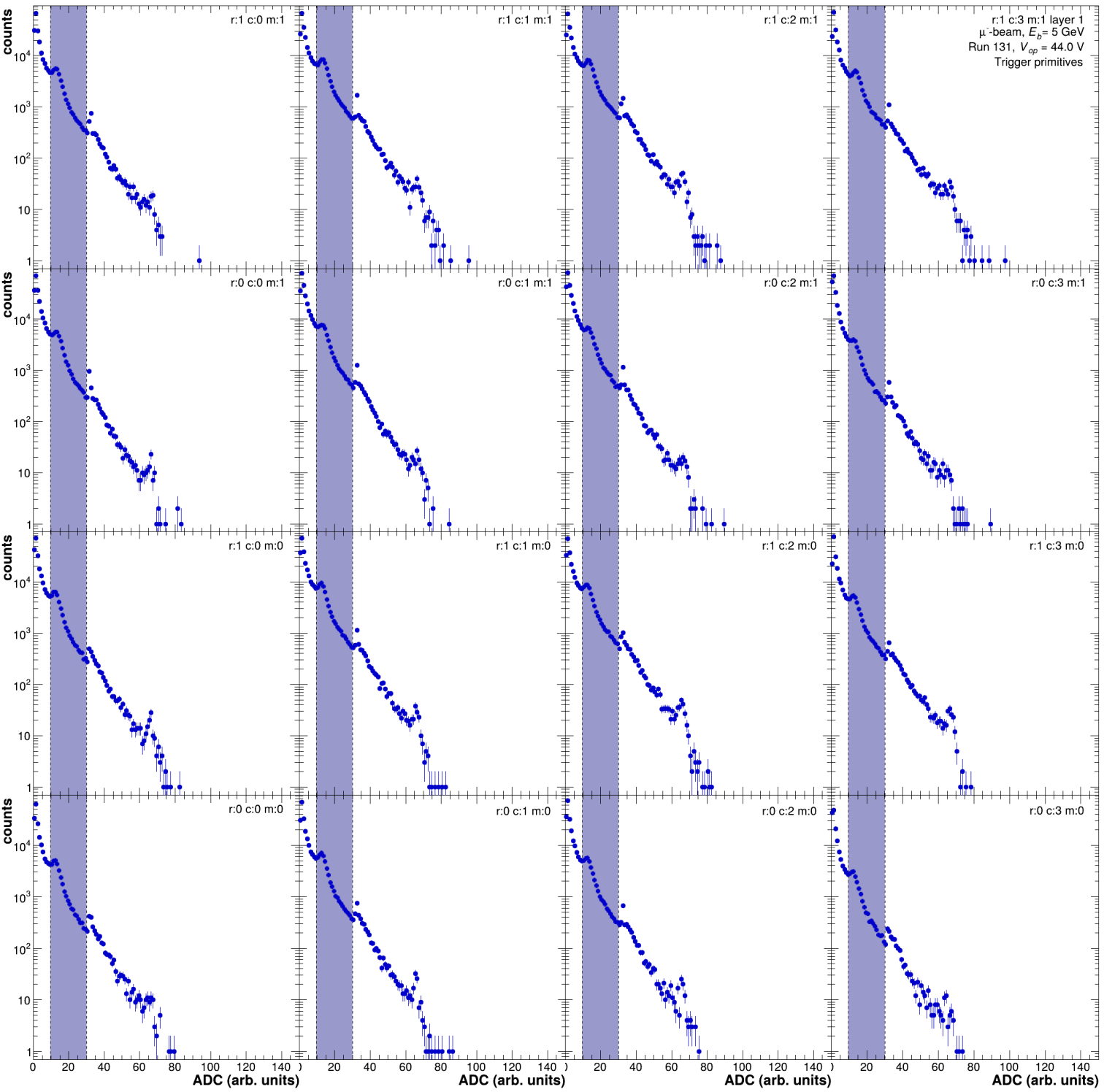

evt. collected = 950631
μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts

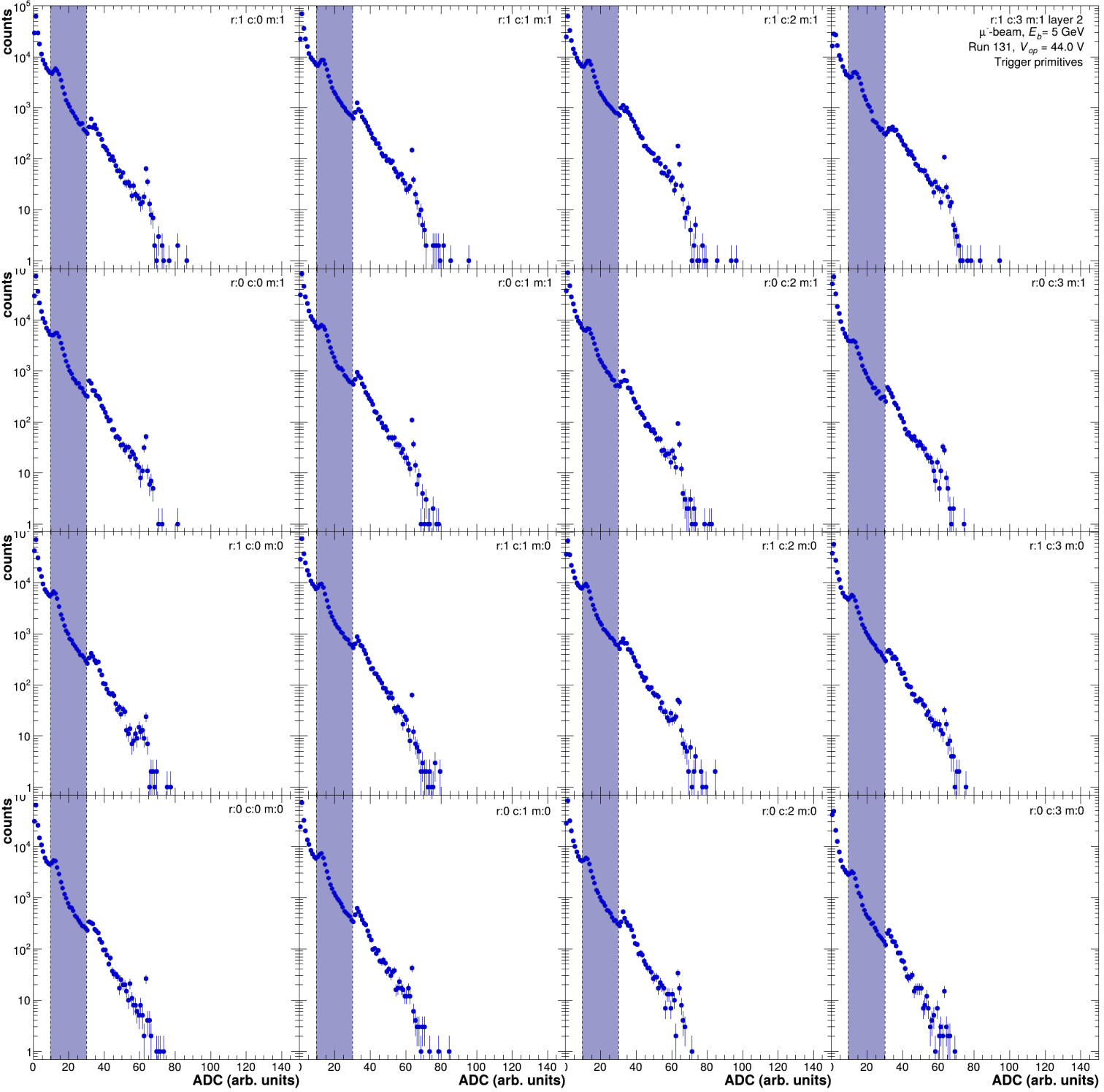

μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
5
10
15
20
25
30
35

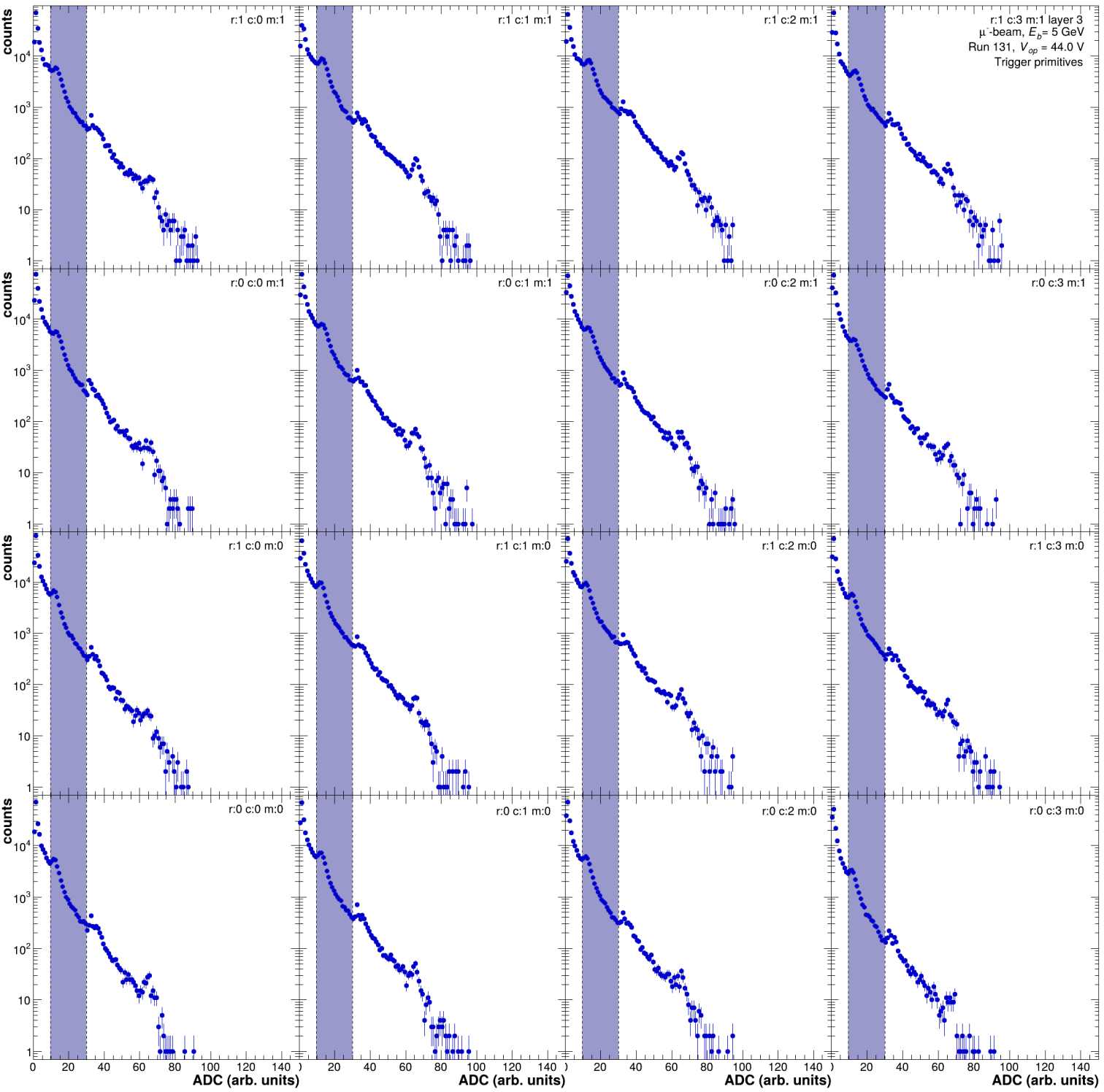

⟨ S/B noise ⟩ = 0.024
μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14

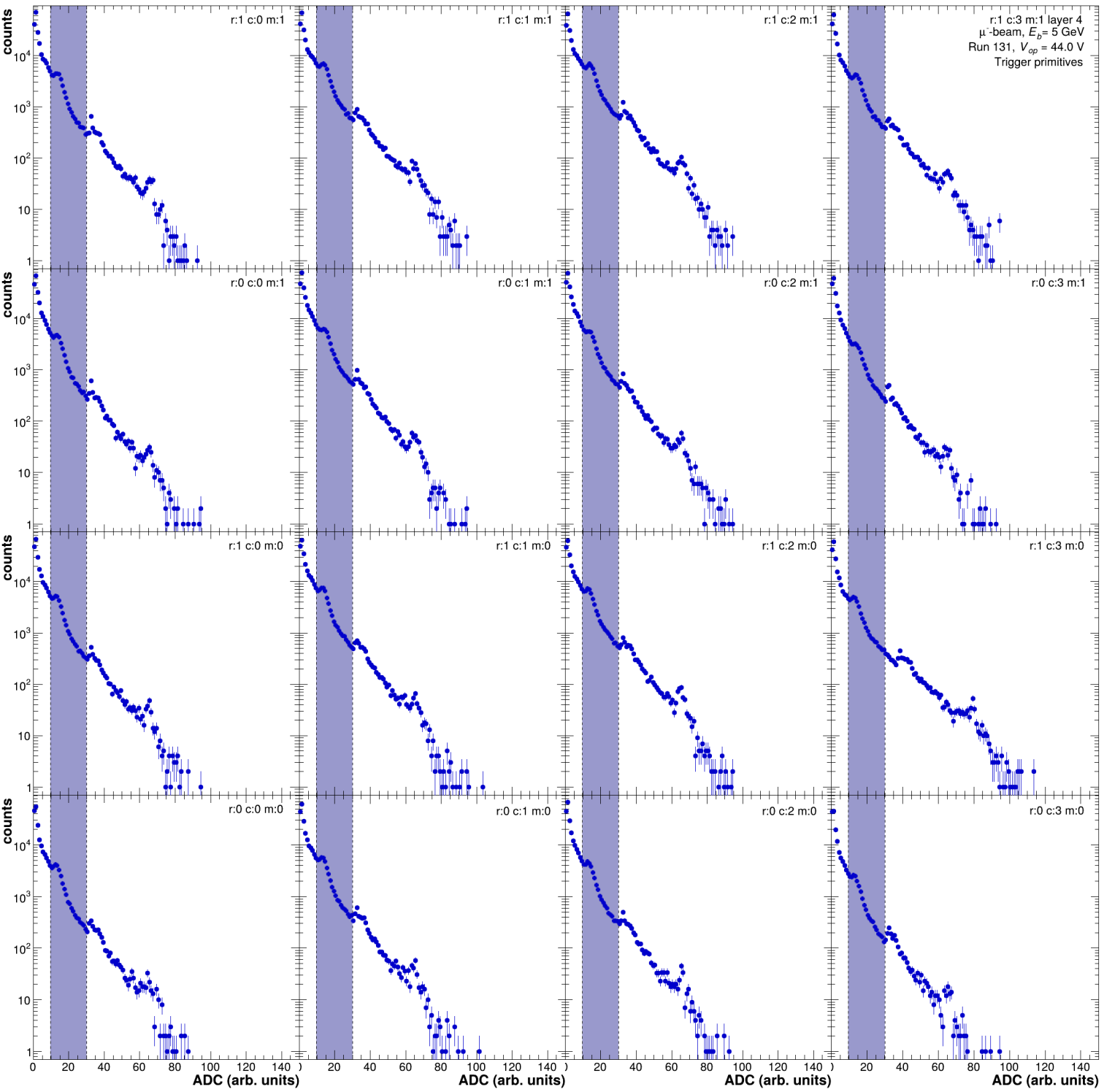

⟨ S/B signal ⟩ = 0.194
μ⁻-beam, E_b= 5 GeV, Run 131, V_{op} = 44.0 V
brd channel
layer
S/B signal region
10^{-1}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

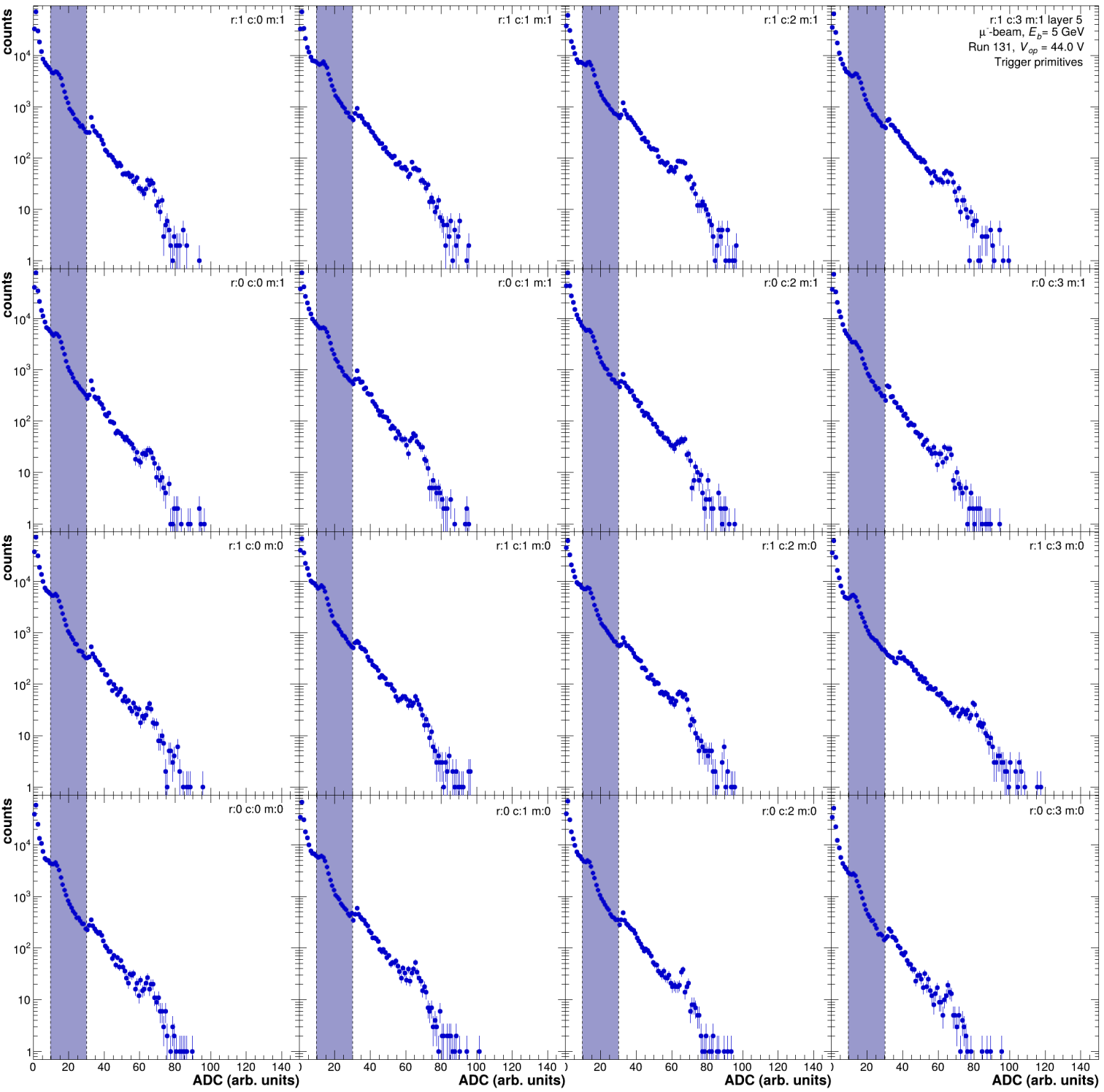

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

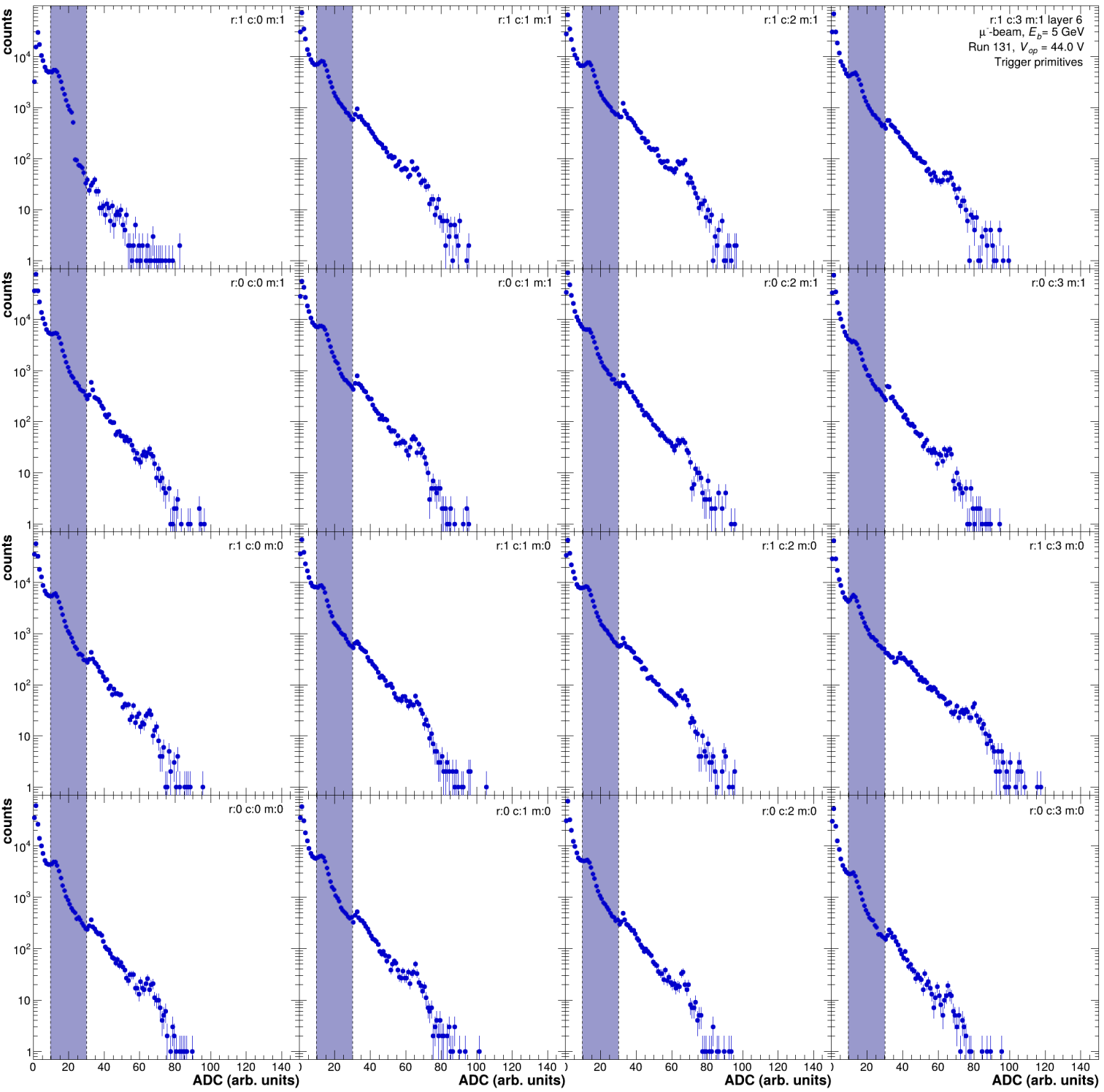

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

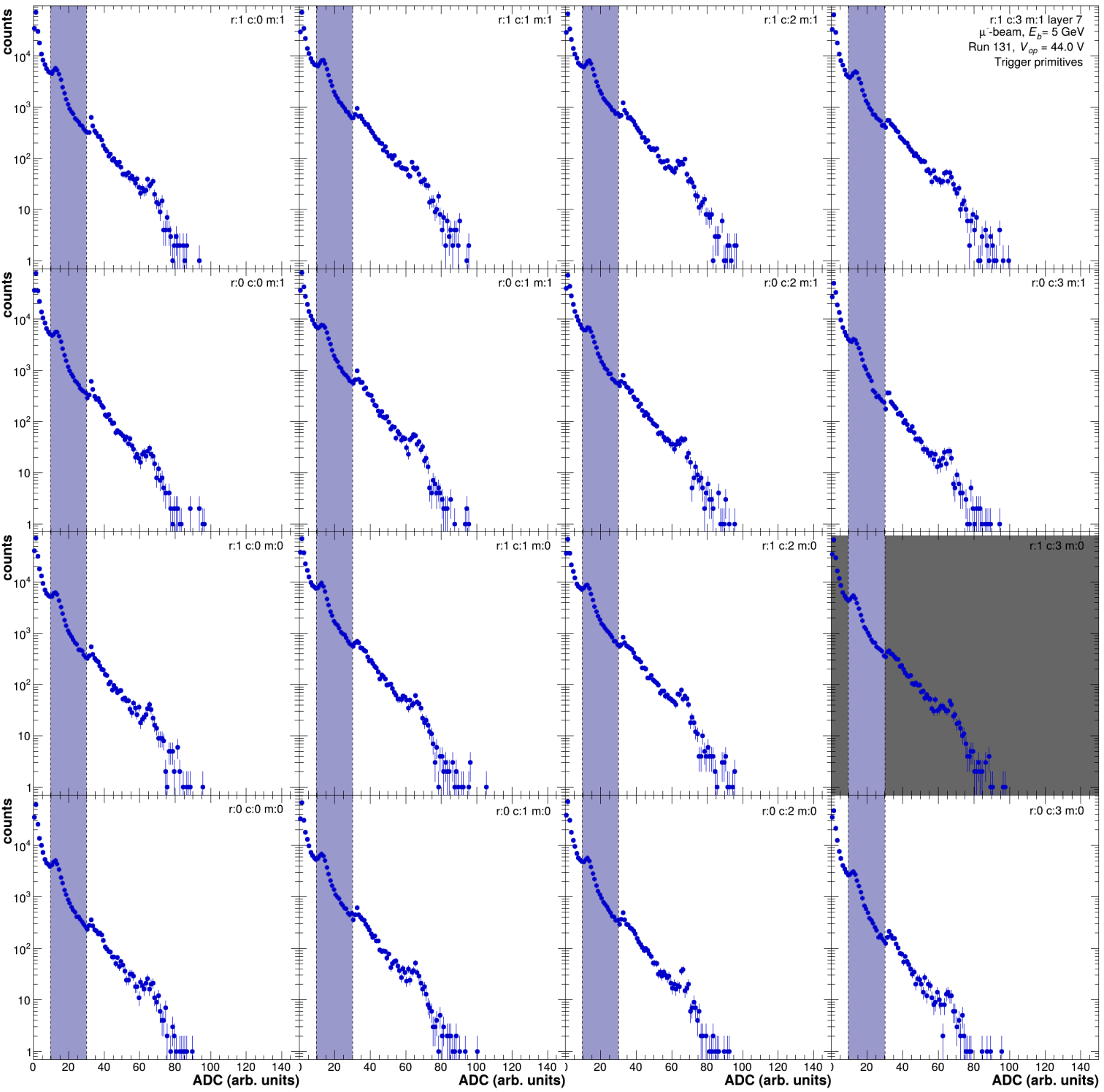

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 131, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)